I YEAR I SEMESTER

S.NO.	COURSE CODE	TITLE OF THE COURSE	SCHEME OF INSTRUCTIION				SCHEME OF EXAMINATION			NO. OF CREDITS
			L/D	T	P	Total	CIA	SEA	Total	
1	23A1100201	Linear Algebra & Calculus	3	0	0	3	30	70	100	3
2	23A1100203	Chemistry	3	0	0	3	30	70	100	3
3	23A1101301	Basic Civil & Mechanical Engineering	3	0	0	3	30	70	100	3
4	23A1105301	Introduction to Programming	3	0	0	3	30	70	100	3
5	23A1103301	Engineering Graphics	1	0	4	5	30	70	100	3
6	23A1105391	Computer Programming Lab	0	0	3	3	30	70	100	1.5
7	23A1112391	IT Workshop	0	0	2	2	30	70	100	1
8	23A1100292	Chemistry Lab	0	0	2	2	30	70	100	1
9	23A1103391	Engineering Workshop	0	0	3	3	30	70	100	1.5
10	23A1100992	Health and wellness, Yoga and sports	0	1	0	1	0	100*	100	0.5
11	23A1100801	Environmental Science	2	0	0	2	30	70*	100	0
TOTAL			15	1	14	30	300	800	1100	20.5

*: No External Evaluation

I YEAR II SEMESTER

S.NO.	COURSE CODE	TITLE OF THE COURSE	SCHEME OF INSTRUCTIION				SCHEME OF EXAMINATION			NO. OF CREDITS
			L	T	P/D	Total	CIA	SEA	Total	
1	BS&H	Communicative English	2	0	0	2	30	70	100	2
2	BS&H	Differential Equations &Vector Calculus	3	0	0	3	30	70	100	3
3	BS&H	Engineering Physics	2	0	2	4	30	70	100	3
4	Engineering Science	Basics of Electrical & Electronics Engineering	3	0	0	3	30	70	100	3
5	Program Core	Data Structures	3	0	0	3	30	70	100	3
6	BS&H	Communicative English Lab	0	0	2	2	30	70	100	1
7	BS&H	Engineering Physics Lab	0	0	2	2	30	70	100	1
8	Engineering Science	Electrical & Electronics Engineering Workshop	0	0	3	3	30	70	100	1.5

9	Program Core Lab	Data Structures Lab	0	0	3	3	30	70	100	1.5
10	Audit	NSS/NCC/Scouts & Guides/Community Service	0	1	0	1	0	100 *	100	0.5
TOTAL			13	1	12	26	270	730	1000	19.5
***: No External Evaluation**										

B.Tech.– II Year I Semester

S.NO.	CATEGORY/ COURSE CODE	TITLE OF THE COURSE	SCHEME OF INSTRUCTIION				SCHEME OF EXAMINATION			NO. OF CREDITS
			L/D	T	P	Total	CIA	SEA	Total	
1	BS&H	Discrete Mathematics & Graph Theory	3	0	0	3	30	70	100	3
2	BS&H	Universal Human Values– Understanding Harmony	2	1	0	3	30	70	100	3
3	Engineering Science	Digital Logic &Computer Organization	3	0	0	3	30	70	100	3
4	Professional Core	Database Management Systems	3	0	0	3	30	70	100	3
5	Professional Core	Object Oriented Programming Through Java	3	0	0	3	30	70	100	3
6	Professional Core	Database Management Systems Lab	0	0	3	3	30	70	100	1.5
7	Professional Core	Object Oriented Programming Through Java Lab	0	0	3	3	30	70	100	1.5
8	Skill Enhancement Course	Python Programming	0	1	2	3	30	70	100	2
9	Audit Course	Constitution of India	2	0	0	2	30	70*	100	0
TOTAL			16	2	8	26	270	630	900	20

*: No External Evaluation

B.Tech.– II Year II Semester

S.NO.	CATEGORY/ COURSE CODE	TITLE OF THE COURSE	SCHEME OF INSTRUCTIION				SCHEME OF EXAMINATION			NO. OF CREDITS
			L/D	T	P	Total	CIA	SEA	Total	
1	Management Course- I	Optimization Techniques	2	0	0	2	30	70	100	2
2	Engineering Science/ Basic Science	Probability & Statistics	3	0	0	3	30	70	100	3
3	Professional Core	Operating Systems	3	0	0	3	30	70	100	3
4	Professional Core	Advanced Data Structures & Algorithm Analysis	3	0	0	3	30	70	100	3
5	Professional Core	Artificial Intelligence	3	0	0	3	30	70	100	3
6	Professional Core	Operating Systems Lab	0	0	3	3	30	70	100	1.5
7	Professional Core	Advanced Data Structures and Algorithm Analysis Lab	0	0	3	3	30	70	100	1.5

8	Skill Enhancement Course	Full Stack Development –I	0	1	2	3	30	70	100	2
9	BS&H	Design Thinking &Innovation	1	0	2	3	30	70	100	2
TOTAL			15	1	10	26	270	630	900	21
Mandatory Community Service Project Internship of 08 weeks duration during summer vacation										

B.Tech. – III Year I Semester

S.No	Category	Title	L	T	P	Credits
1	Professional Core	Machine Learning	3	0	0	3
2	Professional Core	Computer Networks	3	0	0	3
3	Professional Core	Automata Theory and Compiler Design	3	0	0	3
4	Professional Elective-I	1. Object Oriented Analysis and Design 2. Data Warehousing and Data Mining 3. Software Engineering with Agile Methodology 4. DevOps and MLOps 5. 12-week MOOC Swayam/ NPTEL course recommended by the BoS	3	0	0	3
5	Open Elective-I		3	0	0	3
6	Professional Core	Machine Learning Lab	0	0	3	1.5
7	Professional Core	Computer Networks Lab	0	0	3	1.5
8	Skill Enhancement course	Full Stack development-2	0	1	2	2
9	Engineering Science	User Interface Design / SWAYAM Plus - Android Application Development	0	0	2	1
10	Evaluation of Community Service Internship		-	-	-	2
Total			15	1	10	23

S. No.	Category	Title	L	T	P	Credits
1	Professional Core	Deep Learning	3	0	0	3
2	Professional Core	Quantum Computing	3	0	0	3
3	Professional Core	Cryptography & Network Security	3	0	0	3
4	Professional Elective-II	1. Software Testing Methodologies 2. Agentic Systems and Generative AI 3. Cloud Computing 4. Soft Computing 5. 12 week MOOC Swayam/NPTEL course recommended by the BoS	3	0	0	3
5	Professional Elective-III	1. Block Chain Technologies 2. Natural Language Processing 3. Big Data Analytics 4. Distributed Operating System 5. 12 week MOOC Swayam/NPTEL course recommended by the BoS	3	0	0	3
6	Open Elective – II		3	0	0	3
7	Professional Core	Deep Learning Lab	0	0	3	1.5
8	Professional Core	Quantum Algorithms Lab	0	0	3	1.5
9	Skill Enhancement course	Soft skills / / SWAYAM Plus - 21st Century Employability Skills	0	1	2	2
10	Audit Course	Technical Paper Writing &IPR	2	0	0	-
Total			**20**	**1**	**08**	**23**
Mandatory Industry Internship / Mini Project of 08 weeks duration during summer vacation						

Course Code- Machine Learning

Lecture – Tutorial-Practical::	3-0-0	**Internal Marks:**	30
Credits:	3	**External Marks:**	70

Prerequisites:

Course Objectives:

The main objectives of the course are to

- To define machine learning and distinguish between its types, such as supervised and unsupervised learning.
- To understand the applications and use-cases of different machine learning approaches in real-world scenarios.
- To apply supervised learning algorithms, including decision trees and k-nearest neighbors, for classification and regression tasks.
- To implement unsupervised learning techniques like K-means clustering for data grouping and pattern discovery.
- To build foundational skills in selecting and evaluating appropriate ML models for various types of data.

Course Outcomes:		**BTL LEVEL**
Upon successful completion of the course, the student will be able to:		
CO1	Apply ML pipeline from data preprocessing to model evaluation for real-world problems.	L3
CO2	Analyze proximity measures to implement and optimize KNN algorithms for classification/regression.	L4
CO3	Evaluate decision trees, random forests, and Naive Bayes classifiers for different scenarios.	L5
CO4	Create ML solutions using SVMs, neural networks, and kernel methods for complex problems.	L6
CO5	Explain clustering techniques (K-means, hierarchical, fuzzy) and their comparative advantages.	L2

Contribution of Course Outcomes towards the achievement of Program Outcomes (1 – Low, 2-Medium, 3 – High)

	PO 1	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PSO1**	**PSO 2**	**PSO3**
CO1	3	2	-		3	-	-	-	-	-	-	2	3	-	2
CO2	3	3	-	2	3	-	-	-	-	-	-	2	3	-	2
CO3	2	3	2	3	2	-	-	-	-	-	-	3	3	2	2
CO4	3	3	3	2	3	-	-	-	-	-	-	2	3	3	3
CO5	2	-	-		2	-	-	-	-	-	-	3	2	-	2

Syllabus		
UnitNo	**Contents**	**MappedCO**

I	**UNIT-I: Introduction to Machine Learning:** Evolution of Machine Learning, Paradigms for ML, Learning by Rote, Learning by Induction, Reinforcement Learning, Types of Data, Matching, Stages in Machine Learning, Data Acquisition, Feature Engineering, Data Representation, Model Selection, Model Learning, Model Evaluation, Model Prediction, Search and Learning, Data Sets.	**CO1**
II	**UNIT-II: Nearest Neighbor-Based Models:** Introduction to Proximity Measures, Distance Measures, Non-Metric Similarity Functions, Proximity Between Binary Patterns, Different Classification Algorithms Based on the Distance Measures, K-Nearest Neighbor Classifier, Radius Distance Nearest Neighbor Algorithm, KNN Regression, Performance of Classifiers, Performance of Regression Algorithms.	**CO2**
III	**UNIT-III: Models Based on Decision Trees:** Decision Trees for Classification, Impurity Measures, Properties, Regression Based on Decision Trees, Bias–Variance Trade-off, Random Forests for Classification and Regression. The Bayes Classifier: Introduction to the Bayes Classifier, Bayes' Rule and Inference, The Bayes Classifier and its Optimality, Multi-Class Classification, Class Conditional Independence and Naive Bayes Classifier (NBC)	**CO3**
IV	**UNIT-IV: Linear Discriminants for Machine Learning:** Introduction to Linear Discriminants, Linear Discriminants for Classification, Perceptron Classifier, Perceptron Learning Algorithm, Support Vector Machines, Linearly Non-Separable Case, Non-linear SVM, Kernel Trick, Logistic Regression, Linear Regression, Multi-Layer Perceptrons (MLPs), Backpropagation for Training an MLP.	**CO4**
V	**UNIT-V: Clustering:** Introduction to Clustering, Partitioning of Data, Matrix Factorization, Clustering of Patterns, Divisive Clustering, Agglomerative Clustering, Partitional Clustering, K-Means Clustering, Soft Partitioning, Soft Clustering, Fuzzy C-Means Clustering, Rough Clustering, Rough K-Means Clustering Algorithm, Expectation Maximization-Based Clustering, Spectral Clustering.	**CO5**

Text Books:

1. "Machine Learning Theory and Practice", M N Murthy, V S Ananthanarayana, Universities Press (India), 2024

Reference Books:

1. "Machine Learning", Tom M. Mitchell, McGraw-Hill Publication, 2017
2. "Machine Learning in Action",Peter Harrington, DreamTech
3. Applied Machine Learning, 2nd edition, 2021, M. Gopal
4. "Introduction to Data Mining", Pang-Ning Tan, Michel Stenbach, Vipin Kumar, 7th Edition, 2019.

Web-Resources:

https://onlinecourses.nptel.ac.in/noc23_cs18/preview

Course Code- Computer Networks

Lecture – Tutorial-Practical::	3-0-0	**Internal Marks:**	30
Credits:	3	**External Marks:**	70

Prerequisites:

Course Objectives:

The main objectives of the course are to

- To provide insight about networks, topologies, and the key concepts.
- To gain comprehensive knowledge about the layered communication architectures (OSI and TCP/IP) and its functionalities.
- To understand the principles, key protocols, design issues, and significance of each layers in ISO and TCP/IP.
- To know the basic concepts of network services and various network applications.

Course Outcomes:		**BTL LEVEL**
Upon successful completion of the course, the student will be able to:		
CO1	Explain OSI/TCP-IP models and network topologies for system design.	L2
CO2	Implement error detection and flow control mechanisms in network protocols.	L3
CO3	Analyze CSMA and Ethernet protocols for network collision handling.	L4
CO4	Evaluate IP routing algorithms and addressing schemes for network optimization.	L5
CO5	Design complete network applications using TCP/UDP and application protocols.	L6

Contribution of Course Outcomes towards the achievement of Program Outcomes (1 – Low, 2-Medium, 3 – High)

	PO 1	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PSO1**	**PSO 2**	**PSO3**
CO1	2	-	-	-	-	-	-	-	-	-	-	2	2	-	-
CO2	3	2	-	-	2	-	-	-	-	-	-	-	3	-	-
CO3	3	3	-	-	3	-	-	-	-	-	-	2	3	-	-
CO4	3	3	3	-	3	-	-	-	-	-	-	2	3	2	2
CO5	3	3	3	-	3	-	-	-	-	-	-	-	3	3	2

Syllabus		
UnitNo	**Contents**	**MappedCO**
I	**UNIT I: Introduction:** Network Types, LAN, MAN, WAN, Network Topologies Reference models- The OSI Reference Model- the TCP/IP Reference Model - A Comparison of the OSI and TCP/IP Reference Models, OSI Vs TCP/IP. **Physical Layer –**Introduction to Guided Media- Twisted-pair cable, Coaxial cable and Fiber optic cable and introduction about unguided media.	**CO1**
II	**UNIT II: Data link layer:** Design issues, **Framing**: fixed size framing, variable size framing, flow control, error control, error detection and correction codes, CRC, Checksum: idea, one's complement internet checksum, services provided to Network Layer, **Elementary Data Link Layer protocols**: simplex protocol, Simplex stop and wait, Simplex protocol for Noisy Channel.	**CO2**

	Sliding window protocol: One bit, Go back N, Selective repeat-Stop and wait protocol, Data link layer in HDLC, Point to point protocol (PPP)	
III	**UNIT – III: Media Access Control: Random Access**: ALOHA, Carrier sense multiple access (CSMA), CSMA with Collision Detection, CSMA with Collision Avoidance, **Controlled Access:** Reservation, Polling, Token Passing, **Channelization:** frequency division multiple Access(FDMA), time division multiple access(TDMA), code division multiple access(CDMA). **Wired LANs:** Ethernet, Ethernet Protocol, Standard Ethernet, Fast Ethernet(100 Mbps), Gigabit Ethernet, 10 Gigabit Ethernet.	**CO3**
IV	**UNIT – IV: The Network Layer Design Issues** – Store and Forward Packet Switching-Services Provided to the Transport layer- Implementation of Connectionless Service-Implementation of Connection Oriented Service-Comparison of Virtual Circuit and Datagram Networks, Routing Algorithms-The Optimality principle-Shortest path, Flooding, Distance vector, Link state, Hierarchical, Congestion Control algorithms-General principles of congestion control, Congestion prevention polices, Approaches to Congestion Control-Traffic Aware Routing- Admission Control-Traffic Throttling-Load Shedding. Traffic Control Algorithm-Leaky bucket & Token bucket. **Internet Working:** How networks differ- How networks can be connected-Tunnelling, internetwork routing-, Fragmentation, network layer in the internet – IP protocols-IP Version 4 protocol-IPV4 Header Format, IP addresses, Class full Addressing, CIDR, Subnets-IP Version 6-The main IPV6 header, Transition from IPV4 to IPV6, Comparison of IPV4 & IPV6.	**CO4**
V	**UNIT –V: The Transport Layer:** Transport layer protocols: Introduction-services-port number-User data gram protocol-User datagram-UDP services-UDP applications-Transmission control protocol: TCP services- TCP features- Segment-A TCP connection- windows in TCP- flow control-Error control, Congestion control in TCP. **Application Layer** –- World Wide Web: HTTP, Electronic mail-Architecture- web based mail- email security- TELENET-local versus remote Logging-Domain Name System.	**CO5**

Text Books:

1. Computer Networksm, Andrew S Tanenbaum, Fifth Edition. Pearson Education/PHI
2. Data Communications and Networks, Behrouz A. Forouzan, Fifth Edition TMH.

Reference Books:

Data Communications and Networks- Achut S Godbole, AtulKahate

Computer Networks, Mayank Dave, CENGAGE

Web-Resources:

https://onlinecourses.nptel.ac.in/noc22_cs19/preview

Course Code- Automata Theory & Compiler Design

Lecture – Tutorial-Practical::	3-0-0	**Internal Marks:**	30
Credits:	3	**External Marks:**	70

Prerequisites:

Course Objectives:

The main objectives of the course are

- To introduce the fundamental concepts of formal languages, grammars, and their classification in computational theory.
- To design appropriate grammars, finite automata (FAs), and pushdown automata (PDAs) for recognizing different classes of languages.
- To familiarize students with the underlying theoretical foundations and methodologies used in compiler design and construction.
- To introduce advanced parsing techniques, code optimization strategies, and efficient code generation methods for compiler implementation.

Course Outcomes:		**BTL LEVEL**
Upon successful completion of the course, the student will be able to:		
CO1	**Apply** regular expressions and finite automata (DFA/NFA) to design lexical patterns and recognize tokens in programming languages.	L3
CO2	**Analyze** context-free grammars and pushdown automata to handle nested structures and parsing requirements in language design.	L4
CO3	**Implement** lexical analyzers and recursive descent parsers for processing source code tokens and syntax analysis.	L3
CO4	**Evaluate** LR parsing techniques and syntax-directed translation to build robust parser generators and semantic analyzers.	L5
CO5	**Create** intermediate code generators and optimization algorithms for producing efficient target machine code.	L6

Contribution of Course Outcomes towards the achievement of Program Outcomes (1 – Low, 2-Medium, 3 – High)

	PO 1	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PSO1**	**PSO 2**	**PSO3**
CO1	3	-	-	-	2	-	-	-	-	-	-	2	3	-	2
CO2	3	2	-	-	2	-	-	-	-	-	-	2	3	-	2
CO3	3	2	2	-	3	-	-	-	2	-	-	-	3	2	2
CO4	3	3	3	2	3	-	-	-	2	2	2	2	3	3	2
CO5	3	3	3	-	3	-	-	-	2	-	2	-	3	3	3

Syllabus		
UnitNo	**Contents**	**MappedCO**
I	**UNIT – I: Regular Expressions, Languages and Finite Automata** Formal Languages and the Chomsky Hierarchy, Regular Expressions and Regular Languages, Algebraic Laws for Regular Expressions, Applications of Regular Expressions, Abstract model of Finite Automaton, Transition Tables and Transition Graphs, Deterministic Finite Automata (DFA), Nondeterministic Finite Automata (NFA), Converting NFA to DFA, Finite Automata with ϵ transitions (NFA-ϵ), Converting NFA-ϵ to NFA/DFA, Minimization of Finite Automata, Equivalence of FA and Regular Expressions	**CO1**
II	**UNIT-II: Context Free Grammars and Push Down Automata:** Context Free Grammars (CFG) and Context Free Languages (CFL), Design of CFGs, Leftmost and Rightmost Derivations, Parse Trees, Applications of CFGs, Ambiguity in Grammars and Languages, Push Down Automata (PDA), The Language of a PDA, Equivalence of PDAs and CFGs	**CO2**
III	**UNIT-III: Lexical Analysis and Top-Down Parsing** The structure of a compiler, Role of lexical analyzer, Input Buffering, Specification of tokens, Recognition of tokens, The Lexical Analyser Generator –LEX Introduction to Syntax Analysis, Eliminating ambiguity and left recursion from a CFG, Recursive Decent Parsing, LL(1) Grammars, Nonrecursive Predictive Parsing	**CO3**
IV	**UNIT-IV: Bottom-Up Parsing and Syntax Directed Translation** Shift-Reduce Parsing, Simple LR parsing, Canonical LR(1) Parsing, LALR Parsing, Parser Generators Syntax Directed Definitions, Evaluation Orders for SDDs, Syntax Directed Translation Schemes	**CO4**
V	**UNIT-V: Intermediate Code Generation, Code Generation and Optimization:** Three address code, Types and Declarations, Translation of Expressions, Type Checking, Control Flow, Issues in the design of a Code Generator, The Target Language, A simple Code GeneratorBasic Blocks and Flow Graphs, Optimization of Basic Blocks, Peephole Optimization	**CO5**

Text Books:

1. Introduction to Automata Theory, Languages and Computation, J.E.Hopcroft, R.Motwani and J.D.Ullman, 3rd Edition, Pearson, 2008.
2. Compilers Principles, Techniques and Tools, 2nd Edition, Alfred V.Aho, Monica S. Lam, Ravi Sethi, Jeffrey D. Ullman, Pearson

Reference Books:

1. Introduction to Languages and The Theory of Computation, John C. Martin, McGraw Hill.
2. Theory of Computer Science-Automata, Languages and Computation, K.L.P.Mishra and N.Chandrasekaran, 3rd Edition, PHI, 2007
3. Compiler Construction‖, K.V.N. Sunitha, Pearson, 2013
4. Compiler Design, SandeepSaxena, Rajkumar Singh Rathore, S.Chand publication

Web-Resources:

https://onlinecourses.nptel.ac.in/noc25_cs13/preview
https://online.stanford.edu/courses/soe-ycscs1-compilers
https://ocw.mit.edu/courses/18-404j-theory-of-computation-fall-2020/
https://www.tpointtech.com/lex
https://www.classcentral.com/subject/automata
https://mitpress.mit.edu/9780262048637/automata-theory/
https://stackoverflow.com/questions/985837/compiler-design-resources

Course Code- Data Warehousing & Data Mining
(Professional Elective-I)

Lecture – Tutorial-Practical::	3-0-0	**Internal Marks:**	30
Credits:	3	**External Marks:**	70

Prerequisites: Data Structures, Algorithms, Probability & Statistics, Data Base Management Systems

Course Objectives:

The main objectives of the course are to

- Introduce basic concepts and techniques of data warehousing and data mining
- Examine the types of the data to be mined and apply pre-processing methods on raw data
- Discover interesting patterns, analyze supervised and unsupervised models and estimate the accuracy of the algorithms.

Course Outcomes:		**BTL LEVEL**
Upon successful completion of the course, the student will be able to:		
CO1	**Understand** data warehouse architecture, OLAP operations, and cloud data warehouse concepts to design analytical systems for business intelligence.	L2
CO2	**Apply** data cleaning, integration, reduction, and transformation techniques to prepare quality datasets for mining operations.	L3
CO3	**Analyze** decision tree induction, Bayesian classification, and rule-based methods to build predictive models with proper evaluation metrics.	L4
CO4	**Evaluate** frequent itemset generation algorithms (Apriori, FP-Growth) and association rule mining techniques for market basket analysis.	L5
CO5	**Create** clustering solutions using K-means, hierarchical clustering, and DBSCAN algorithms for unsupervised pattern discovery and data segmentation.	L6

Contribution of Course Outcomes towards the achievement of Program Outcomes (1 – Low, 2-Medium, 3 – High)

	PO 1	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PSO1**	**PSO 2**	**PSO3**
CO1	2	-	-	-	2	-	-	-	-	-	-	2	2	-	-

CO2	3	2	-	-	3	-	-	-	2	-	-	-	3	-	2
CO3	3	3	2	2	3	-	-	-	2	2	-	2	3	2	2
CO4	3	3	2	3	3	2	-	-	2	2	2	2	3	3	2
CO5	3	3	3	-	3	-	-	-	3	-	2	-	3	3	2

Syllabus		
UnitNo	**Contents**	**MappedCO**
I	UNIT–I: Data Warehousing and Online Analytical Processing: Basic concepts, Data Warehouse Modeling: Data Cube and OLAP, Data Warehouse Design and Usage, Data Warehouse Implementation, Cloud Data Warehouse, Data Mining and Patten Mining, Technologies, Applications, Major issues, Data Objects & Attribute Types, Basic Statistical Descriptions of Data, Data Visualization, Measuring Data Similarity and Dissimilarity.	**CO1**
II	**UNIT II: Data Preprocessing:** An Overview, Data Cleaning, Data Integration, Data Reduction, Data Transformation and Data Discretization.	**CO2**
III	**UNIT–III: Classification:** Basic Concepts, General Approach to solving a classification problem, Decision Tree Induction: Attribute Selection Measures, Tree Pruning, Scalability and Decision Tree Induction, Visual Mining for Decision Tree Induction, Bayesian Classification Methods: Bayes Theorem, Naïve Bayes Classification, Rule-Based Classification, Model Evaluation and Selection.	**CO3**
IV	**UNIT–IV: Association Analysis:** Problem Definition, Frequent Itemset Generation, Rule Generation: Confident Based Pruning, Rule Generation in Apriori Algorithm, Compact Representation of frequent item sets, FP-Growth Algorithm.	**CO4**
V	**UNIT–V: Cluster Analysis:** Overview, Basics and Importance of Cluster Analysis, Clustering techniques, Different Types of Clusters; K-means: The Basic K-means Algorithm, K-means Additional Issues, Bi-secting K Means, Agglomerative Hierarchical Clustering: Basic Agglomerative Hierarchical Clustering Algorithm DBSCAN: Traditional Density Center-Based Approach, DBSCAN Algorithm, Strengths and Weaknesses.	**CO5**

Text Books:

1. Data Mining concepts and Techniques, 3rd edition, Jiawei Han, Michel Kamber, Elsevier, 2011.
2. Introduction to Data Mining: Pang-Ning Tan & Michael Steinbach, Vipin Kumar, Pearson, 2012.

Reference Books:

1. Data Mining: VikramPudi and P. Radha Krishna, Oxford Publisher.
2. Data Mining Techniques, Arun K Pujari, 3rd edition, Universities Press,2013.

Web-Resources:

https://www.edx.org/learn/data-mining
https://www.edx.org/learn/data-warehouse
https://www.geeksforgeeks.org/data-mining/
https://www.tutorialspoint.com/weka/index.htm
https://onlinecourses.nptel.ac.in/noc21_cs06/preview

Course Code- Object Oriented Analysis and Design
(Professional Elective-I)

Lecture – Tutorial-Practical::	3-0-0	**Internal Marks:**	30
Credits:	3	**External Marks:**	70

Prerequisites: Object-Oriented Programming (OOP) Concepts, Mathematical & Logical Foundation, Data Structures

Course Objectives:
The main objectives of the course are

To introduce the principles of complex systems design and the inherent complexity challenges in software development.

To develop proficiency in UML modeling principles, object-oriented modeling concepts, and software development lifecycle integration.

To design structural models using class diagrams, object diagrams, and advanced structural modeling techniques for software architecture.

To implement behavioral modeling techniques including use case diagrams, interaction diagrams, and activity diagrams for system requirements analysis.

To apply advanced behavioral modeling and architectural modeling concepts using state machines, component diagrams, and deployment diagrams for comprehensive system design.

Course Outcomes:		**BTL LEVEL**
Upon successful completion of the course, the student will be able to:		
CO1	**Understand** data warehouse architecture, OLAP operations, and cloud data warehouse concepts to design analytical systems for business intelligence.	L2
CO2	**Apply** data cleaning, integration, reduction, and transformation techniques to prepare quality datasets for mining operations.	L3
CO3	**Analyze** decision tree induction, Bayesian classification, and rule-based methods to build predictive models with proper evaluation metrics.	L4
CO4	**Evaluate** frequent itemset generation algorithms (Apriori, FP-Growth) and association rule mining techniques for market basket analysis.	L5
CO5	**Create** clustering solutions using K-means, hierarchical clustering, and DBSCAN algorithms for unsupervised pattern discovery and data segmentation.	L6

Contribution of Course Outcomes towards the achievement of Program Outcomes (1 – Low, 2-Medium, 3 – High)

	PO 1	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PSO1**	**PSO 2**	**PSO3**

CO1	2	2	-	-	2	-	-	-	-	-	-	2	2	-	2
CO2	3	2	2	-	3	-	-	1	2	2	-	2	3	2	2
CO3	3	2	3	-	3	-	-	1	2	2	2	-	3	2	2
CO4	3	3	2	2	3	1	-	2	2	3	2	2	3	2	2
CO5	3	3	3	3	3	1	-	2	3	2	3	-	3	3	2

Syllabus		
UnitNo	**Contents**	**MappedCO**
I	UNIT I: **Introduction:** The Structure of Complex systems, The Inherent Complexity of Software, Attributes of Complex System, Organized and Disorganized Complexity, Bringing Order to Chaos, Designing Complex Systems. **Case Study:** System Architecture: Satellite-Based Navigation	**CO1**
II	UNIT II: **Introduction to UML:** Importance of modeling, principles of modeling, object oriented modeling, conceptual model of the UML, Architecture, and Software Development Life Cycle. **Basic Structural Modeling:** Classes, Relationships, common Mechanisms, and diagrams. **Case Study:** Control System: Traffic Management.	**CO2**
III	UNIT III: **Class & Object Diagrams:** Terms, concepts, modeling techniques for Class & Object Diagrams. **Advanced Structural Modeling:** Advanced classes, advanced relationships, Interfaces, Types and Roles, Packages. **Case Study:** AI: Cryptanalysis.	**CO3**
IV	UNIT IV: **Basic Behavioral Modeling-I:** Interactions, Interaction diagrams Use cases, Use case Diagrams, Activity Diagrams. **Case Study:** Web Application: Vacation Tracking System	**CO4**
V	UNIT V: **Advanced Behavioral Modeling:** Events and signals, state machines, processes and Threads, time and space, state chart diagrams. **Architectural Modeling:** Component, Deployment, Component diagrams and Deployment diagrams. **Case Study:** Weather Forecasting	**CO5**

Text Books:

1. Grady BOOCH, Robert A. Maksimchuk, Michael W. ENGLE, Bobbi J. Young, Jim Conallen, Kellia Houston , “Object- Oriented Analysis and Design with Applications”, 3rd edition, 2013, PEARSON.
2. Grady Booch, James Rumbaugh, Ivar Jacobson: The Unified Modeling Language User Guide, Pearson Education.

Reference Books:

1. Meilir Page-Jones: Fundamentals of Object Oriented Design in UML, Pearson Education.
2. Pascal Roques: Modeling Software Systems Using UML2, WILEY- Dreamtech India Pvt. Ltd.
3. Atul Kahate: Object Oriented Analysis & Design, The McGraw-Hill Companies.
4. Appling UML and Patterns: An introduction to Object – Oriented Analysis and Design and Unified Process, Craig Larman, Pearson Education

Web-Resources:

https://onlinecourses.nptel.ac.in/noc25_cs52/preview
https://onlinecourses.nptel.ac.in/noc24_cs119/preview
https://www.geeksforgeeks.org/unified-modeling-language-uml-
https://www.tutorialspoint.com/uml/index.htm
https://www.javatpoint.com/uml
https://www.uml.org/
https://www.visual-paradigm.com/guide/uml-unified-modeling-language/uml-class-diagram-tutorial/
https://www.geeksforgeeks.org/unified-modeling-language-uml-sequence-diagrams/

Course Code- DevOps and MLOps
(Professional Elective-I)

Lecture – Tutorial-Practical::	3-0-0	**Internal Marks:**	30
Credits:	3	**External Marks:**	70

Prerequisites: NIL

Course Objectives:

The main objectives of the course are

To understand DevOps and MLOps fundamentals, principles, and their application in software and machine learning development lifecycles.

To develop skills in data management, versioning, automation, and CI/CD pipeline implementation for ML workflows.

To master model deployment strategies, production monitoring, and infrastructure management using industry-standard tools and frameworks.

Course Outcomes:		**BTL LEVEL**
Upon successful completion of the course, the student will be able to:		
CO1	Analyze DevOps and MLOps concepts, compare with DataOps, and evaluate CI/CD principles for software versus ML pipelines	L4
CO2	Implement data versioning using Git, DVC, MLflow and apply experiment tracking for model reproducibility.	L3
CO3	Create automated CI/CD pipelines using GitHub Actions, Jenkins, and implement scalable training with distributed frameworks.	L6

CO4	Evaluate deployment strategies (Blue-Green, Canary, A/B Testing) and implement monitoring systems for drift detection.	L5
CO5	Create containerized ML infrastructure using Docker, Kubernetes, Kubeflow and assess security compliance in ML systems.	L6

Contribution of Course Outcomes towards the achievement of Program Outcomes (1 – Low, 2-Medium, 3 – High)

	PO 1	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PSO1**	**PSO 2**	**PSO3**
CO1	2	-	-	-	2	-	-	-	-	2	2	2	2	-	2
CO2	2	2	-	-	2	-	-	-	-	-	-	2	2	2	2
CO3	3	2	2	-	3	-	-	-	-	-	2	-	3	2	2
CO4	3	2	2	-	2	2	-	2	2	2	2	2	3	2	3
CO5	3	3	2	2	3	2	2	2	3	2	2	-	3	3	2

Syllabus		
UnitNo	**Contents**	**MappedCO**
I	**UNIT-I** Introduction to DevOps and MLOps: Understanding DevOps and MLOps, Importance of DevOps in software development, Overview of MLOps and its role in machine learning, Difference between DevOps, MLOps, and DataOps, Key principles and best practices, CI/CD in traditional software vs. ML pipelines, Model development to deployment lifecycle	**CO1**
II	Unit 2: Data Management and Versioning in MLOps: Managing data and model versioning (Git, DVC, MLflow), Handling large datasets for ML, Data versioning tools: DVC, Pachyderm, Integrating data quality checks into workflows, Experiment tracking and model reproducibility (MLflow, TensorBoard	**CO2**
III	Unit 3: CI/CD and Automation in MLOps: Automation in DevOps & MLOps, Automating ML model training processes, Scalable training techniques (distributed training with Horovod, Ray), CI/CD for Machine Learning and DevOps, GitHub Actions, Jenkins, GitLab CI/CD for automated pipelines, Continuous training (CT) for model retraining, Evaluation metrics & validation strategies	**CO3**
IV	Unit 4: Model Deployment and Production Monitoring: DevOps & MLOps Deployment Strategies: Blue-Green, Canary, A/B Testing, Online, batch, and streaming deployments, Model Serving and Scaling: TensorFlow Serving, TorchServe, FastAPI, Model optimization techniques (Quantization, ONNX), Monitoring & Model Performance Management: Prometheus & Grafana, Detecting model drift, data drift, and concept drift, Strategies for retraining and updating ML models	**CO4**
V	Unit 5: Infrastructure, Orchestration & Security: Containerization & Orchestration: Docker, Kubernetes, Kubeflow, KServe, Managing compute resources with Kubeflow, Feature Store and Workflow Orchestration: Feast, Tecton, Apache Airflow, Kubeflow Pipelines, Security & Compliance in DevOps & MLOps, DevSecOps: Security in CI/CD pipelines, Model security risks & GDPR compliance	**CO5**

Text Books:

1. Joyner, Joseph., Devops for Beginners: Devops Software Development Method Guide for Software Developers and It Professionals, 1st Edition Mihails Konoplows, 2015.
2. Alisson Machado de Menezes., Hands-on DevOps with Linux,1st Edition, BPB Publications, India, 2021.
3. "Practical MLOps: Operationalizing Machine Learning Models", Noah Gift, Alfredo Deza, O'Reilly Media, 1st Edition, 2021.
4. "Machine Learning with Kubernetes: Deploy, Manage, and Scale Models with Kubernetes", Oleg Iegorov, Faisal Masood, Krishnaram Kenthapadi, O'Reilly Media, 1st Edition, 2021.
5. "Designing Machine Learning Systems: An Iterative Process for Production-Ready Applications", Chip Huyen, O'Reilly Media, 1st Edition, 2022.
6. The DevOps Handbook: How to Create World-Class Agility, Reliability, & Security in Technology Organizations" Gene Kim, Jez Humble, Patrick Debois, and John Willis, IT Revolution Press, 2nd Edition (2021).

Reference Books:

1. Len Bass, Ingo Weber, Liming Zhu. DevOps: A Software Architect's Perspective. Addison Wesley; ISBN-10
2. Gene Kim Je Humble, Patrick Debois, John Willis. The DevOps Handbook, 1st Edition, IT Revolution Press, 2016.
3. Verona, Joakim Practical DevOps, 1st Edition, Packt Publishing, 2016.
4. Joakim Verona. Practical Devops, Ingram short title; 2nd edition (2018). ISBN10: 1788392574
5. Deepak Gaikwad, Viral Thakkar. DevOps Tools from Practitioner's Viewpoint. Wiley publications. ISBN: 9788126579952

Web-Resources:

https://github.com/milanm/DevOps-Roadmap
https://www.coursera.org/courses?query=devops
https://labs.play-with-docker.com/
https://labs.play-with-k8s.com/
https://infyspringboard.onwingspan.com/web/en/page/lex_auth_01366558964890009633

Course Code- Software Engineering with Agile Methodology
(Professional Elective-I)

Lecture – Tutorial-Practical::	3-0-0	**Internal Marks:**	30
Credits:	3	**External Marks:**	70

Prerequisites: NIL

Course Objectives:

To cover the underlying concepts and techniques used in Software Engineering and to understand the modern way of Software development using Agile methodology.

Course Outcomes:	**BTL LEVEL**
Upon successful completion of the course, the student will be able to:	

CO1	Analyze various Software Development Life Cycle models and requirements engineering processes to select appropriate methodologies for different project contexts	L4
CO2	Apply software design principles, UML modeling techniques, and comprehensive testing strategies to develop robust software systems	L3
CO3	Evaluate software maintenance strategies, configuration management practices, and project management approaches for effective software project execution.	L5
CO4	Analyze Agile methodology fundamentals and compare Agile versus traditional development approaches in modern software development contexts.	L4
CO5	Create comprehensive Agile-based software solutions using multiple frameworks and implement effective agile project delivery strategies.	L6

Contribution of Course Outcomes towards the achievement of Program Outcomes (1 – Low, 2-Medium, 3 – High)

	PO 1	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PSO1**	**PSO 2**	**PSO3**
CO1	2	2	-	-	-	2	-	2	-	2	2	3	2	-	2
CO2	3	3	2	-	2	2	-	3	2	3	3	2	3	3	-
CO3	3	2	3	-	2	2	-	2	3	2	2	2	2	3	3
CO4	3	3	2	3	2	3	-	3	2	3	2	3	3	2	2
CO5	3	3	2	3	3	2	2	2	2	3	3	2	3	-	3

Syllabus		
UnitNo	**Contents**	**MappedCO**
I	**Unit I: Introduction to Software Engineering** Software Development Life Cycles, SDLC Models: Waterfall, V-Model, Prototype Model, Incremental, Evolutionary, RAD, Spiral. Project Planning, Metrics for Project Size Estimation, Project Estimation Techniques, Requirements Gathering and Analysis, Software Requirements Specification (SRS). Software Product and Process Characteristics, Software Process Models, Evolutionary Process Models and Agile processes. Software Process customization and improvement, CMM, Product and Process Metrics, Functional and Non-functional requirements, Requirement Sources and Elicitation Techniques, Analysis Modeling for Function-oriented and Object-oriented software development, Use case Modeling, System and Software Requirement Specifications, Requirement Validation, Traceability	**CO1**
II	**Unit II: Software Design, Analysis and Testing** The Software Design Process, Design Concepts and Principles, Software Modeling and UML, Architectural Design, Architectural Views and Styles, User Interface Design, Function oriented Design, SA/SD Component Based Design, Design Metrics. Software Static and Dynamic analysis, Code inspections, Software Testing,	**CO2**

	Fundamentals, Software Test Process, Testing Levels, Test Criteria, Test Case Design, Test Oracles, Test Techniques, Testing Frameworks, Test Plan, Test Metrics, Testing Tools.	
III	**Unit-III: Software Maintenance & Software Project Management** Need and Types of Maintenance, Software Configuration Management (SCM), Software Change Management, Version Control, Change control and Reporting, Program Comprehension Techniques, Re-engineering, Reverse Engineering, Tool Support. Project Management Concepts, Feasibility Analysis, Project and Process Planning, Resources Allocations, Software efforts, Schedule, and Cost estimations, Project Scheduling and Tracking, Risk Assessment and Mitigation, Software Quality Assurance (SQA). Project Plan, Project Metrics.	**CO3**
IV	**Unit-IV: Fundamentals of Agile Methodology** Introduction to Agile software development methodology, Life Cycle of Agile development, Agile v/s Traditional software development(Waterfall model)Agile Manifesto: Principles, Benefits and Challenges of Agile, Agile Values, Agile Model, Phases of Agile Model.	**CO4**
V	**Unit-V: Software Development using Agile Methodology** Gathering requirement using Agile way, User Stories: The currency of agile development, Characteristics of good user stories, Generating User Stories, Agile estimation and planning, Implementation of agile, Applying an Agile Mindset to a Project, Roles in agile development, Agile Frameworks: Scrum, Kanban, Crystal, XP, ASD, DSDM.	**CO5**

Text Books:

1. Fundamentals of Software Engineering, RajibMall,5thEdition, PHI.
2. Software Engineering a Practitioner's Approach, Roger S. Pressman, 9thEdition, McGraw Hill International Edition.

Reference Books:

1. Software Engineering, IanSommerville,10thEdition, Pearson.
2. Software Engineering, Principles and Practices, Deepak Jain, Oxford University Press.

Web-Resources:

1. https://nptel.ac.in/courses/106/105/106105182/
2. https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_01260589506387148827_shared/overview
3. https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_01338269041100390473 5_shared/overview

Course Code- MACHINE LEARNING LAB

Lecture – Tutorial-Practical::	0-0-3	**Internal Marks:**	30
Credits:	1.5	**External Marks:**	70

Prerequisites: Linear Algebra, Statistics and Probability, Calculus, Python Programming Fundamentals

Course Objectives:

The objectives of this course are to introduce

- To learn about computing central tendency measures and Data preprocessing techniques
- To learn about classification and regression algorithms
- To apply different clustering algorithms for a problem.

Course Outcomes:		**BTL LEVEL**
Upon successful completion of the course, the student will be able to:		
CO1	**Apply** central tendency measures and **implement** data preprocessing techniques including attribute selection, missing value handling, discretization, and outlier elimination.	L3
CO2	**Implement** KNN algorithm, **construct** decision tree algorithms, and **apply** Random Forest with parameter optimization.	L3
CO3	**Demonstrate** Naïve Bayes classification, **apply** SVM, and **implement** linear and logistic regression.	L3
CO4	**Analyze** Multi-layer Perceptron architecture and **evaluate** performance through systematic experimentation.	L4
CO5	**Analyze** K-means clustering performance and **evaluate** Fuzzy C-Means and Expectation Maximization techniques.	L4

Contribution of Course Outcomes towards the achievement of Program Outcomes (1 – Low, 2-Medium, 3 – High)

	PO 1	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PSO1**	**PSO 2**	**PSO3**
CO1	3	2	-	-	3	-	-	2	-	2	-	2	3	-	2
CO2	3	3	2	-	3	-	-	2	2	2	2	2	3	3	2
CO3	3	3	2	-	3	2	-	2	2	2	2	2	2	3	3
CO4	3	3	2	2	3	2	-	3	3	3	2	3	3	2	2
CO5	3	3	2	2	3	-	-	2	2	3	2	3	3	-	3

List of Experiments:

1. Compute Central Tendency Measures: Mean, Median, Mode Measure of Dispersion: Variance, Standard Deviation.
2. Apply the following Pre-processing techniques for a given dataset.
    a. Attribute selection
    b. Handling Missing Values
    c. Discretization
    d. Elimination of Outliers
3. Apply KNN algorithm for classification and regression
4. Demonstrate decision tree algorithm for a classification problem and perform parameter tuning for better results
5. Demonstrate decision tree algorithm for a regression problem
6. Apply Random Forest algorithm for classification and regression
7. Demonstrate Naïve Bayes Classification algorithm.
8. Apply Support Vector algorithm for classification
9. Demonstrate simple linear regression algorithm for a regression problem
10. Apply Logistic regression algorithm for a classification problem
11. Demonstrate Multi-layer Perceptron algorithm for a classification problem
12. Implement the K-means algorithm and apply it to the data you selected. Evaluate performance by measuring the sum of the Euclidean distance of each example from its class center. Test the performance of the algorithm as a function of the parameters K.
13. Demonstrate the use of Fuzzy C-Means Clustering
14. Demonstrate the use of Expectation Maximization based clustering algorithm

Text Books:

1. "Machine Learning Theory and Practice", M N Murthy, V S Ananthanarayana, Universities Press (India), 2024

Reference Books:

1. "Machine Learning", Tom M. Mitchell, McGraw-Hill Publication, 2017
2. "Machine Learning in Action",Peter Harrington, DreamTech
3. Applied Machine Learning, 2nd edition, 2021, M. Gopal
4. "Introduction to Data Mining", Pang-Ning Tan, Michel Stenbach, Vipin Kumar, 7th Edition, 2019.

Web-Resources:

https://onlinecourses.nptel.ac.in/noc23_cs18/preview

Course Code- Computer Networks Lab

Lecture – Tutorial-Practical::	0-0-3	**Internal Marks:**	30
Credits:	1.5	**External Marks:**	70

Prerequisites: Computer Networks Theory Foundation, Programming Fundamentals, Mathematics and Logic

Course Objectives:

The objectives of this course are to introduce

- **To** learn basic concepts of computer networking and acquire practical understanding of network protocols with emphasis on TCP/IP protocol suite.
- **To** provide hands-on experience with network devices and Ethernet/Internet networking by assembling networks and conducting experiments.
- **To** understand the layered network architecture through practical implementation of data link layer protocols and error detection mechanisms.
- **To** implement and analyze network routing algorithms and congestion control mechanisms for efficient data transmission.
- **To** gain proficiency in network analysis tools like Wireshark, Nmap, and NS2 simulator for network monitoring, security assessment, and performance evaluation.

Course Outcomes:		**BTL LEVEL**
Upon successful completion of the course, the student will be able to:		
CO1	**Apply** knowledge of network devices and LAN connections, and **implement** data link layer framing methods including character stuffing, bit stuffing, and checksum techniques.	L3
CO2	**Implement** Hamming code generation for error detection and correction, and **apply** CRC polynomials (CRC-12, CRC-16, CRC-CCIP) for data integrity verification.	L3
CO3	**Analyze** sliding window protocols including Go-Back-N and Selective Repeat, **evaluate** Stop and Wait protocol performance, and **implement** congestion control using leaky bucket algorithm.	L4
CO4	**Analyze** Dijkstra's algorithm for shortest path computation, **evaluate** distance vector routing algorithm performance, and **implement** broadcast tree algorithms for network topology management.	L4
CO5	**Evaluate** network traffic using Wireshark packet analysis, **assess** network security using Nmap scanning techniques, and **create** network simulations using NS2 simulator to analyze packet drops, congestion, and throughput performance.	L5

Contribution of Course Outcomes towards the achievement of Program Outcomes (1 – Low, 2-Medium, 3 – High)

	PO 1	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PSO1**	**PSO 2**	**PSO3**
CO1	3	2	2	-	2	-	-	2	2	2	2	2	2	2	2
CO2	3	3	2	-	2	-	-	2	2	2	2	2	3	2	2
CO3	3	3	2	2	3	-	-	2	2	3	2	3	3	3	3
CO4	3	3	2	2	3	-	-	2	3	3	3	2	3	2	3
CO5	3	3	3	3	3	2	-	3	3	3	2	3	3	3	3

List of Experiments:

1. Study of Network devices in detail and connect the computers in Local Area Network.
2. Write a Program to implement the data link layer farming methods such as
   i) Character stuffing ii) bit stuffing.
3. Write a Program to implement data link layer farming method checksum.
4. Write a program for Hamming Code generation for error detection and correction.
5. Write a Program to implement on a data set of characters the three CRC polynomials – CRC 12, CRC 16 and CRC CCIP.
6. Write a Program to implement Sliding window protocol for Goback N.
7. Write a Program to implement Sliding window protocol for Selective repeat.
8. Write a Program to implement Stop and Wait Protocol.
9. Write a program for congestion control using leaky bucket algorithm
10. Write a Program to implement Dijkstra‘s algorithm to compute the Shortest path through a graph.
11. Write a Program to implement Distance vector routing algorithm by obtaining routing table at each node (Take an example subnet graph with weights indicating delay between nodes).
12. Write a Program to implement Broadcast tree by taking subnet of hosts.
13. Wireshark
    i. Packet Capture Using Wire shark
    ii. Starting Wire shark
    iii. Viewing Captured Traffic
    iv. Analysis and Statistics & Filters.
14. How to run Nmap scan
15. Operating System Detection using Nmap
16. Do the following using NS2 Simulator
    i. NS2 Simulator-Introduction
    ii. Simulate to Find the Number of Packets Dropped
    iii. Simulate to Find the Number of Packets Dropped by TCP/UDP
    iv. Simulate to Find the Number of Packets Dropped due to Congestion
    v. Simulate to Compare Data Rate& Throughput

Text Books:

1. Computer Networksm, Andrew S Tanenbaum, Fifth Edition. Pearson Education/PHI
2. Data Communications and Networks, Behrouz A. Forouzan, Fifth Edition TMH.

Reference Books:

Data Communications and Networks- Achut S Godbole, AtulKahate

Computer Networks, Mayank Dave, CENGAGE

Web-Resources:

https://onlinecourses.nptel.ac.in/noc22_cs19/preview

Course Code- Full Stack Development - 2
(Skill Enhancement Course)

Lecture – Tutorial-Practical::	0-0-3	**Internal Marks:**	30
Credits:	1.5	**External Marks:**	70

Prerequisites: Programming Fundamentals, Web Development Fundamentals, Database Concepts

Course Objectives:
The main objectives of the course are to

- Make use of router, template engine and authentication using sessions to develop application in ExpressJS.
- Build a single page application using RESTful APIs in ExpressJS
- Apply router and hooksin designing ReactJS application
- Make use of MongoDB queries to perform CRUD operations on document database

Course Outcomes:		**BTL LEVEL**
Upon successful completion of the course, the student will be able to:		
CO1	**Implement** ExpressJS routing with HTTP methods and middleware, **apply** templating engines for dynamic content rendering, and **create** form data handling mechanisms for web applications.	L3
CO2	**Analyze** session management using cookies and authentication mechanisms, **evaluate** database connectivity with MongoDB using Mongoose, and **design** RESTful APIs for single-page applications.	L4
CO3	**Apply** JSX syntax for rendering HTML elements, **implement** functional and class-based components with props and states, and **create** event-driven user interfaces with styling capabilities.	L3
CO4	**Analyze** conditional rendering and list rendering techniques, **evaluate** React Router for navigation, and **design** React forms with hooks for state management and component data sharing.	L4
CO5	**Evaluate** MongoDB installation and configuration processes, **assess** CRUD operations with database collections and records, and **create** complete full-stack applications integrating MERN stack technologies.	L5

Contribution of Course Outcomes towards the achievement of Program Outcomes (1 – Low, 2-Medium, 3 – High)

	PO 1	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PSO1**	**PSO 2**	**PSO3**
CO1	3	2	3	-	2	-	-	2	2	2	-	2	2	3	2
CO2	3	3	3	-	3	2	-	3	2	2	2	2	2	3	3
CO3	3	2	3	-	2	-	-	2	2	2	-	2	-	3	2
CO4	3	3	3	-	3	-	-	2	2	2	2	2	2	3	3
CO5	3	3	3	-	3	2	-	3	3	3	3	3	3	3	3

List of Experiments:

- ExpressJS – Routing, HTTP Methods, Middleware, Templating, Form Data
- ExpressJS – Cookies, Sessions, Authentication, Database, RESTful APIs
- ReactJS – Render HTML, JSX, Components – function & Class, Props and States, Styles, Respond to Events
- ReactJS – Conditional Rendering, Rendering Lists, React Forms, React Router, Updating the Screen
- ReactJS – Hooks, Sharing data between Components, Applications – To-do list and Quiz
- MongoDB – Installation, Configuration, CRUD operations, Databases, Collections and Records

Sample Experiments:

1. ExpressJS – Routing, HTTP Methods, Middleware.

a. Write a program to define a route, Handling Routes, Route Parameters, Query Parameters and URL building.
b. Write a program to accept data, retrieve data and delete a specified resource using http methods.
c. Write a program to show the working of middleware.

2. ExpressJS – Templating, Form Data

a. Write a program using templating engine.
b. Write a program to work with form data.

3. ExpressJS – Cookies, Sessions, Authentication

a. Write a program for session management using cookies and sessions.
b. Write a program for user authentication.

4. ExpressJS – Database, RESTful APIs

a. Write a program to connect MongoDB database using Mangoose and perform CRUD operations.
b. Write a program to develop a single page application using RESTful APIs.

5. ReactJS – Render HTML, JSX, Components – function & Class

a. Write a program to render HTML to a web page.
b. Write a program for writing markup with JSX.
c. Write a program for creating and nesting components (function and class).

6. ReactJS – Props and States, Styles, Respond to Events

a. Write a program to work with props and states.
b. Write a program to add styles (CSS & Sass Styling) and display data.
c. Write a program for responding to events.

7. ReactJS – Conditional Rendering, Rendering Lists, React Forms

a. Write a program for conditional rendering.
b. Write a program for rendering lists.
c. Write a program for working with different form fields using react forms.

8. ReactJS – React Router, Updating the Screen

a. Write a program for routing to different pages using react router.
b. Write a program for updating the screen.

9. ReactJS – Hooks, Sharing data between Components

a. Write a program to understand the importance of using hooks.
b. Write a program for sharing data between components.

10. MongoDB – Installation, Configuration, CRUD operations

a. Install MongoDB and configure ATLAS
b. Write MongoDB queries to perform CRUD operations on document using insert(), find(), update(), remove()

11. MongoDB – Databases, Collections and Records

a. Write MongoDB queries to Create and drop databases and collections.
b. Write MongoDB queries to work with records using find(), limit(), sort(), createIndex(), aggregate().

12. Augmented Programs: (Any 2 must be completed)

a. Design a to-do list application using NodeJS and ExpressJS.
b. Design a Quiz app using ReactJS.
c. Complete the MongoDB certification from MongoDB University website.

Text Books:

1. Pro MERN Stack: Full Stack Web App Development with Mongo, Express, React, and Node, Vasan Subramanian, 2nd edition, APress, O'Reilly.
2. Node.Js in Action, Mike Cantelon, Mark Harter, T.J. Holowaychuk, Nathan Rajlich,Manning Publications. (Chapters 1-11)
3. React Quickly, AzatMardan,Manning Publications(Chapters 1-8,12-14)

Web-Resources:

1. ExpressJS - https://www.tutorialspoint.com/expressjs
2. ReactJS - https://www.w3schools.com/REACT (and) https://react.dev/learn#
3. MongoDB - https://learn.mongodb.com/learning-paths/introduction-to-mongodb

Course Code- User Interface Design

Lecture – Tutorial-Practical::	0-0-2	**Internal Marks:**	30
Credits:	1	**External Marks:**	70

Prerequisites: Programming Fundamentals, Mobile Development Concepts, Web Technologies Foundation

Course Objectives:

The main objectives of the course are to

- Learns to Implement Flutter Widgets and Layouts
- Understands Responsive UI Design and with Navigation in Flutter
- Knowledge on Widges and customize widgets for specific UI elements, Themes
- Understand to include animation apart from fetching data

Course Outcomes:		**BTL LEVEL**
Upon successful completion of the course, the student will be able to:		
CO1	**Apply** Flutter and Dart SDK installation procedures, **implement** basic Dart programming concepts, and **create** fundamental Flutter widget structures using Text, Image, Container components.	L3
CO2	**Analyze** different layout structures using Row, Column, and Stack widgets, **evaluate** responsive design principles, and **design** adaptive UI interfaces using media queries and breakpoints.	L4
CO3	**Analyze** navigation patterns between screens using Navigator and named routes, **evaluate** stateful vs stateless widget concepts, and **implement** state management using setState and Provider patterns.	L4
CO4	**Create** custom widgets for specific UI elements, **apply** theming and styling techniques, **implement** form design with validation and error handling mechanisms.	L3
CO5	**Evaluate** Flutter animation framework implementation, **assess** REST API data fetching and display techniques, and **create** comprehensive testing strategies with debugging tools for mobile applications.	L5

Contribution of Course Outcomes towards the achievement of Program Outcomes (1 – Low, 2-Medium, 3 – High)

	PO 1	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PSO1**	**PSO 2**	**PSO3**
CO1	3	2	2	-	2	-	-	2	2	2	-	2	2	3	2
CO2	2	3	3	-	2	2	-	2	2	2	2	2	-	3	3
CO3	3	3	3	-	2	-	-	2	2	2	-	2	2	-	2
CO4	2	3	3	-	2	-	-	2	2	2	2	2	-	3	2
CO5	3	3	3	-	3	2	-	2	3	3	2	3	2	3	3

List of Experiments:

Students need to implement the following experiments

1. a) Install Flutter and Dart SDK.
   b) Write a simple Dart program to understand the language basics.

2. a) Explore various Flutter widgets (Text, Image, Container, etc.).
   b) Implement different layout structures using Row, Column, and Stack widgets.

3. a) Design a responsive UI that adapts to different screen sizes.
   b) Implement media queries and breakpoints for responsiveness.

4. a) Set up navigation between different screens using Navigator.
   b) Implement navigation with named routes.

5. a) Learn about stateful and stateless widgets.
   b) Implement state management using set State and Provider.

6. a) Create custom widgets for specific UI elements.
   b) Apply styling using themes and custom styles.

7. a) Design a form with various input fields.
   b) Implement form validation and error handling.

8. a) Add animations to UI elements using Flutter's animation framework.
   b) Experiment with different types of animations (fade, slide, etc.).

9. a) Fetch data from a REST API.
   b) Display the fetched data in a meaningful way in the UI.

10. a) Write unit tests for UI components.
    b) Use Flutter's debugging tools to identify and fix issues.

Text Books:

1. Marco L. Napoli, Beginning Flutter: A Hands-on Guide to App Development.
2. Rap Payne, Beginning App Development with Flutter: Create Cross-Platform Mobile Apps 1st Edition, Apres
3. Richard Rose, Flutter & Dart Cookbook, Developing Full stack Applications for the Cloud, Oreilly.

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B. Tech III Year – II Semester

Course Code- DEEP LEARNING

Lecture – Tutorial-Practical::	3-0-0	**Internal Marks:**	30
Credits:	3	**External Marks:**	70

Prerequisites: Mathematical foundation in linear algebra, calculus, probability, and statistics; proficiency in Python programming with object-oriented concepts basic understanding of machine learning algorithms and optimization techniques; familiarity with numerical computing libraries (NumPy, Pandas)

Course Objectives:

The main objectives of the course are

- **To** understand the fundamentals of biological and artificial neural networks, including basic computational units and perceptron learning algorithms.
- **To** comprehend the architecture and training mechanisms of feedforward and deep neural networks, including backpropagation and optimization techniques.
- **To** learn advanced training methodologies and regularization techniques for improving neural network performance and preventing overfitting.
- **To** explore specialized neural network architectures including recurrent, convolutional, and generative models for various applications.
- **To** understand recent advances in deep learning including transformers, variational autoencoders, and their applications in vision, NLP, and speech processing.

Course Outcomes:		**BTL LEVEL**
Upon successful completion of the course, the student will be able to:		
CO1	**Apply** biological neuron concepts to McCulloch-Pitts computational units and linear perceptron models with threshold logic and convergence analysis	L3
CO2	**Analyze** multilayer perceptron architectures using gradient descent, backpropagation algorithms, empirical risk minimization, regularization, and autoencoder structures.	L4
CO3	**Evaluate** modern optimization algorithms including Adam, RMSprop, Adagrad, second-order methods, regularization techniques, and saddle point problems.	L4
CO4	**Create** recurrent neural networks with LSTM and GRU, convolutional networks, and generative models using MCMC sampling techniques.	L5
CO5	**Assess** variational autoencoders, transformer architectures, attention mechanisms, and GPT models for vision, NLP, and speech applications.	L5

Contribution of Course Outcomes towards the achievement of Program Outcomes (1 – Low, 2-Medium, 3 – High)

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO1	PSO 2	PSO3
CO1	3	2	2		2	-	-	-	-	2	-	2	3	-	-
CO2	3	3	3	2	2	-	-	-	-	2	-	2	3	-	2
CO3	3	3	2	3	2	-	-	-	-	2	-	3	3	-	2
CO4	3	3	3	2	3	-	-	-	-	3	-	2	3	-	3
CO5	3	-	-		3	-	-	-	-	-	-	3	2	2	2

Syllabus		
UnitNo	**Contents**	**MappedCO**
I	**UNIT-I:** Basics- Biological Neuron, Idea of computational units, McCulloch–Pitts unit and Thres holding logic, Linear Perceptron, Perceptron Learning Algorithm, Linear separability, Convergence theorem for Perceptron Learning Algorithm.	**CO1**
II	UNIT-II: Feed forward Networks- Multilayer Perceptron, Gradient Descent, Backpropagation, Empirical Risk Minimization, regularization, autoencoders. Deep Neural Networks: Difficulty of training deep neural networks, Greedy layer wise training.	**CO2**
III	UNIT-III: Better Training of Neural Networks –Newer optimization methods for neural networks (Adagrad, adadelta, rmsprop, adam, NAG), second order methods for training, Saddle point proble minneural networks, Regularization methods (dropout, drop connect, batch normalization).	**CO3**
IV	UNITIV: Recurrent Neural Networks- Back propagation through time, Long Short-Term Memory, Gated Recurrent Units, Bidirectional LSTMs, Bidirectional RNNs. Convolutional Neural Networks: LeNet, AlexNet. Generative models: Restrictive Boltzmann Machines (RBMs), Introduction to MCMC and Gibbs Sampling, gradient computations in RBMs, Deep Boltzmann Machines.	**CO4**
V	UNITV: Recent trends- Variational Autoencoders, Transformers, GPT Applications: Vision, NLP, Speech	**CO5**

TextBooks:

1. Deep Learning, Ian Good fellow and Yoshua Bengio and Aaron Courville, MIT Press, 2016

Reference Books:

1. Neural Networks: A Systematic Introduction, RaúlRojas,1996
2. Pattern Recognition and Machine Learning, Christopher Bishop,2007
3. Deep Learning with Python, François Chollet, Manning Publications,2017

Web-Resources:

https://nptel.ac.in/courses/106/106/106106184/
https://nptel.ac.in/courses/106/106/106106139/
https://nptel.ac.in/courses/117/117/117117081/

Course Code- Quantum Computing

Lecture – Tutorial-Practical::	3-0-0	**Internal Marks:**	30
Credits:	3	**External Marks:**	70

Prerequisites:

Engineering Mathematics , Physics , Programming in C/Python, Data Structures and Algorithms, Linear Algebra, Discrete Mathematics, Probability and Statistics; fundamental understanding of classical algorithms and computational complexity

Course Objectives:

The main objectives of the course are to

To introduce the fundamentals of quantum computing, the problem-solving approach using finite dimensional mathematics

Course Outcomes:		**BTL LEVEL**
Upon successful completion of the course, the student will be able to:		
CO1	**Understand** the historical development of quantum computing with mathematical, physical, and biological foundations including classical vs quantum logical operations.	L3
CO2	**Analyze** linear algebra, Hilbert spaces, quantum mechanics principles including superposition, entanglement, measurements, and density operators for quantum systems.	L4
CO3	**Create** quantum circuits using single and multiple qubit gates with Bloch sphere representations and Bell states for information processing.	L4
CO4	**Evaluate** classical vs quantum computational complexity through implementations of Deutsch's, Shor's factorization, and Grover's search algorithms.	L5

CO5	**Assess** quantum error correction methods, fault-tolerant computation, cryptography protocols, and teleportation for secure communication systems.	L5

Contribution of Course Outcomes towards the achievement of Program Outcomes (1 – Low, 2-Medium, 3 – High)

	PO 1	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PSO1**	**PSO 2**	**PSO3**
CO1	3	2	-	-	2	-	-	-	-	2	-	3	3	-	2
CO2	3	3	2	-	2	-	-	-	-	2	-	3	3	-	2
CO3	3	3	3	-	3	-	-	-	-	2	-	3	3	-	3
CO4	3	3	3	2	3	-	-	-	-	3	2	3	3	2	3
CO5	3	3	3	2	3	2	-	2	-	3	2	3	2	2	3

Syllabus		
UnitNo	**Contents**	**MappedCO**
I	**UNIT - I** History of Quantum Computing: Importance of Mathematics, Physics and Biology. Introduction to Quantum Computing: Bits Vs Qubits, Classical Vs Quantum logical operations	**CO1**
II	**UNIT - II** Background Mathematics: Basics of Linear Algebra, Hilbert space, Probabilities and measurements.	**CO2**
III	**UNIT - III** Qubit: Physical implementations of Qubit. Qubit as a quantum unit of information. The Bloch sphere Quantum Circuits: single qubit gates, multiple qubit gates, designing the quantum circuits. Bell states.	**CO3**
IV	**UNIT - IV** Quantum Algorithms: Classical computation on quantum computers. Relationship between quantum and classical complexity classes. Deutsch's algorithm, Deutsch's-Jozsa algorithm, Shor's factorization algorithm, Grover's search algorithm.	**CO4**
V	**UNIT - V** Noise and error correction: Graph states and codes, Quantum error correction, fault-tolerant computation. Quantum Information and Cryptography: Comparison between classical and quantum information theory. Quantum Cryptography, Quantum teleportation	**CO5**

Text Books:

1. Quantum Computation and Quantum Information, Nielsen M. A.,Cambridge
2. Programming Quantum Computers, Essential Algorithms and Code Samples, Eric R Johnson, Nic Harrigan, Mercedes Ginemo, Segovia, Oreilly

Reference Books:

1. Quantum Computing for Computer Scientists, Noson S. Yanofsk, Mirco A. Mannucci
2. Principles of Quantum Computation and Information,Benenti G., Casati G. and Strini G., Vol.I: Basic Concepts, Vol II
3. Basic Tools and Special Topics, World Scientific. Pittenger A. O., An Introduction to Quantum Computing Algorithms

e- Resources:

https://onlinecourses.nptel.ac.in/noc25_cs61/preview

Course Code- Cryptography & Network Security

Lecture – Tutorial-Practical::	3-0-0	**Internal Marks:**	30
Credits:	3	**External Marks:**	70

Prerequisites: Students should have knowledge in

Discrete Mathematics, Mathematics (Algebra & Calculus), Programming Fundamentals, Data Structures and Algorithms, Computer Networks Basics

Course Objectives:

The main objectives of the course are to

- Understand and Security goals, mathematical foundations, cryptographic principles
- Apply and implement Symmetric encryption algorithms (DES, AES), mathematical structures
- Analyze and compare Asymmetric encryption systems (RSA, ECC), comparative analysis
- Evaluate and assess Data integrity, digital signatures, authentication protocols
- Design and develop Comprehensive network security architectures and solutions

Course Outcomes:		**BTL LEVEL**
Upon successful completion of the course, the student will be able to:		
CO1	Apply security principles, attack analysis, and mathematical foundations (modular arithmetic, linear congruence) for cryptographic system analysis	**L3**
CO2	Implement and analyze DES, AES, block ciphers, and stream ciphers using algebraic structures and finite field mathematics	**L4**
CO3	Evaluate RSA, Rabin, ElGamal, and Elliptic Curve Cryptography using prime factorization and Chinese Remainder Theorem	**L5**
CO4	Analyze hash functions (SHA-512, Whirlpool), digital signatures, and key management systems including Kerberos protocols	**L4**
CO5	Design comprehensive security systems integrating PGP/S/MIME, SSL/TLS, IPSec, firewalls, and intrusion detection	**L6**

Contribution of Course Outcomes towards the achievement of Program Outcomes (1 – Low, 2-Medium, 3 – High)

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	-	2	2	2	2	2	2	2	2	2	-	3	2	2

CO2	2	3	3	3	2	-	-	2	-	-	-	2	3	2	2
CO3	3	3	-	-	3	2	2	-	2	-	2	2	2	3	-
CO4	2	-	3	3	3	2	-	-	-	2	-	-	3	3	3
CO5	3	3	3	2	-	3	2	3	-	-	2	2	2	2	3

Syllabus		
UnitNo	**Contents**	**MappedCO**
I	**UNIT I:** **Basic Principles:** Security Goals, Cryptographic Attacks, Services and Mechanisms, Mathematics of Cryptography- integer arithmetic, modular arithmetic, matrices, linear conguence.	**CO1**
II	**UNIT II:** **Symmetric Encryption:** Mathematics of Symmetric Key Cryptography-algebraic structures, GF(2^n) Fields, Introduction to Modern Symmetric Key Ciphers-modern block ciphers, modern stream ciphers, Data Encryption Standard- DES structure, DES analysis, Security of DES, Multiple DES, Advanced Encryption Standard-transformations, key expansions, AES ciphers, Analysis of AES.	**CO2**
III	**UNIT III:** **Asymmetric Encryption:** Mathematics of Asymmetric Key Cryptography-primes, primality testing, factorization, CRT, Asymmetric Key Cryptography- RSA crypto system, Rabin cryptosystem, Elgamal Crypto system, ECC	**CO3**
IV	**UNIT IV:** **Data Integrity, Digital Signature Schemes & Key Management :** Message Integrity and Message Authentication-message integrity, Random Oracle model, Message authentication, Cryptographic Hash Functions-whirlpool, SHA-512, Digital Signature- process, services, attacks, schemes, applications, Key Management-symmetric key distribution, Kerberos.	**CO4**
V	**UNIT V:** **Network Security-I:** Security at application layer: PGP and S/MIME, Security at the Transport Layer: SSL and TLS, **Network Security-II :** Security at the Network Layer: IPSec-two modes, two security protocols, security association, IKE, ISAKMP, System Security-users, trust, trusted systems, buffer overflow, malicious software, worms, viruses, IDS, Firewalls.	**CO5**

Text Books:

1. Cryptography and Network Security, 3rd Edition Behrouz A Forouzan, Deb deep Mukhopadhyay, McGraw Hill,2015
2. Cryptography and Network Security,4th Edition, William Stallings, (6e) Pearson,2006
3. Everyday Cryptography, 1st Edition, Keith M.Martin, Oxford,2016

Reference Books:

1. Network Security and Cryptography, 1st Edition, Bernard Meneges, Cengage Learning,2018

e-Resources : https://onlinecourses.nptel.ac.in/noc22_cs90/preview

Course Code- Software Testing Methodologies
(Professional Elective-II)

Lecture – Tutorial-Practical::	3-0-0	**Internal Marks:**	30
Credits:	3	**External Marks:**	70

Prerequisites: Students should have knowledge in

Programming Fundamentals, Software Engineering**,** Data Structures and Algorithms

Discrete Mathematics**,** Database Management Systems**,** Computer Networks **,** Operating Systems

Course Objectives:
The main objectives of the course are to

- To **define and explain** fundamental software testing concepts including testing processes, bug taxonomies, flow graphs, and the purpose and consequences of software defects in various development contexts.
- To **implement and execute** various testing methodologies including path testing, data flow testing, transaction flow testing, and domain testing techniques to design comprehensive test cases for given software systems.
- **To examine and differentiate** between different testing strategies by analyzing flow graphs, state diagrams, decision tables, and domain boundaries to identify optimal testing approaches for specific software scenarios.
- To **assess and critique** the effectiveness of different testing methodologies by comparing path coverage, data flow adequacy, state transition completeness, and domain testing sufficiency to determine testing strategy quality and coverage gaps.
- **To design and develop** automated testing frameworks and comprehensive testing strategies by integrating multiple testing methodologies, utilizing modern testing tools (Selenium, JMeter, SoapUI), and constructing systematic test management processes for complex software projects.

Course Outcomes:	**BTL LEVEL**
Upon successful completion of the course, the student will be able to:	

CO1	Apply path testing techniques using flow graphs to achieve comprehensive code coverage and systematic test case generation.	**L3**
CO2	Design transaction flow, data flow, and domain testing strategies for complex software validation and interface testing.	**L6**
CO3	Construct path expressions and decision tables using regular expressions and logic-based methods for specification testing.	**L6**
CO4	Model software behavior using state graphs and implement transition testing for behavioral requirement validation.	**L3**
CO5	Apply graph matrix operations and demonstrate proficiency with modern automated testing tools like Selenium/JMeter.	**L3**

Contribution of Course Outcomes towards the achievement of Program Outcomes (1 – Low, 2-Medium, 3 – High)

	PO 1	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PSO1**	**PSO 2**	**PSO3**
CO1	**3**	**3**	**2**	**-**	**-**	-	-	-	-	**-**	-	**2**	**3**	**2**	**2**
CO2	**3**	**3**	**3**	**3**	**2**	**2**	-	**2**	**2**	**-**	-	**2**	**2**	**2**	**2**
CO3	**3**	**3**	**-**	**2**	**-**	**2**	-	-	**2**	**2**	-	**-**	**3**	**2**	**2**
CO4	**3**	**3**	**3**	**-**	**2**	**2**	-	**2**	**2**	**2**	**2**	**2**	**2**	**3**	**2**
CO5	**2**	**2**	**2**	**3**	**3**	-	-	**2**	**2**	**2**	**2**	**-**	**3**	2	**3**

Syllabus		
UnitNo	**Contents**	**Mapped CO**
I	**UNIT - I** Introduction: Purpose of testing, Dichotomies, model for testing, consequences of bugs, taxonomy of bugs Flow graphs and Path testing: Basics concepts of path testing, predicates, path predicates and achievable paths, path sensitizing, path instrumentation, application of path testing.	**CO1**
II	**UNIT - II** Transaction Flow Testing: transaction flows, transaction flow testing techniques. Data Flow testing: Basics of data flow testing, strategies in data flow testing, application of data flow testing. Domain Testing: domains and paths, Nice & ugly domains, domain testing, domains and interfaces testing, domain and interface testing, domains and testability.	**CO2**
III	**UNIT - III** Paths, Path products and Regular expressions: path products & path expression, reduction procedure, applications, regular expressions & flow anomaly detection. Logic Based Testing: overview, decision tables, path expressions, kv charts, specifications.	**CO3**

IV	**UNIT - IV** State, State Graphs and Transition testing: state graphs, good & bad state graphs, state testing, Testability tips.	**CO4**
V	**UNIT - V** Graph Matrices and Application: Motivational overview, matrix of graph, relations, power of a matrix, node reduction algorithm, building tools. (Student should be given an exposure to a tool like Jmeter/selenium/soapUI/Catalon).	**CO5**

TEXT BOOKS

1. Software Testing techniques – Boris Beizer, Dreamtech, second edition. 2. Software Testing Tools – Dr.K.V.K.K.Prasad, Dreamtech.

REFERENCES BOOKS:

1. The craft of software testing – Brian Marick, Pearson Education.
2. Software Testing Techniques – SPD(Oreille)
3. Software Testing in the Real World – Edward Kit, Pearson.
4. Effective methods of Software Testing, Perry, John Wiley.
5. Art of Software Testing – Meyers, John Wiley.**Web-Resources:**

https://onlinecourses.nptel.ac.in/noc25_cs66/preview

Course Code- Soft Computing (Professional Elective-II)

Lecture – Tutorial-Practical::	3-0-0	**Internal Marks:**	30
Credits:	3	**External Marks:**	70

Prerequisites: Students should have knowledge in

Computer Networks, **Operating Systems**, **Database Management Systems**, **Programming Fundamentals**, **Computer Organization and Architecture**

Course Objectives:
The main objectives of the course are

To understand soft computing paradigms, their differences from hard computing, and explore fuzzy logic, neural networks, and evolutionary computing techniques.

To develop skills in fuzzy set theory, evolutionary algorithms, and hybrid soft computing systems for solving complex real-world problems.

To master advanced neuro-fuzzy modeling techniques including ANFIS and adaptive network systems for intelligent decision-making applications.

Course Outcomes:		**BTL LEVEL**
Upon successful completion of the course, the student will be able to:		
CO1	Analyze soft computing vs hard computing concepts and compare fuzzy logic, neural networks, and evolutionary computing applications.	**L4**
CO2	Apply fuzzy set operations, membership functions, and implement fuzzification/defuzzification processes.	**L3**
CO3	Implement evolutionary algorithms (EP, ES) and design evolutionary systems for optimization problems.	**L3**
CO4	Evaluate neuro-fuzzy hybrid systems and analyze genetic neuro-hybrid system properties.	**L5**
CO5	Create ANFIS architectures with hybrid learning algorithms and design adaptive neuro-fuzzy networks	**L6**

Contribution of Course Outcomes towards the achievement of Program Outcomes (1 – Low, 2-Medium, 3 – High)

	PO 1	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PSO1**	**PSO 2**	**PSO3**
CO1	3	3	2	-	2	-	-	-	-	2	-	2	2	2	2
CO2	3	3	3	3	2	2	-	2	2	-	-	-	3	3	2
CO3	3	-	-	-	-	2	-	-	2	2	-	2	-	3	-
CO4	3	3	3	2	2	2	-	2	-	2	2	2	2	3	2
CO5	2	2	2	3	3	-	-	-	2	2	2	3	3	3	-

Syllabus		
UnitNo	**Contents**	**MappedCO**
I	Introduction: Introduction, soft computing vs. hard computing, various types of soft computing techniques, and applications of soft computing. Basic tools of soft computing – Fuzzy logic, neural network, evolutionary computing. Introduction: Neural networks, application scope of neural networks, fuzzy logic, genetic algorithm, and hybrid systems.	**CO1**

II	**UNIT II:** Fuzzy Sets and Logic: Basic concepts of fuzzy logic, Fuzzy sets and Crisp sets, Fuzzy set theory and operations, Properties of fuzzy sets, Fuzzy and Crisp relations, Fuzzy to Crisp conversion. Membership functions, interference in fuzzy logic, fuzzy if-then rules, Fuzzy implications and Fuzzy algorithms, Fuzzyfications and Defuzzifications.	**CO2**
III	**UNIT III:** Evolutionary Computing: Basic Evolutionary Processes, EV: A Simple Evolutionary System, Evolutionary Systems as Problem Solvers, A Historical Perspective, Canonical Evolutionary Algorithms - Evolutionary Programming, Evolution Strategies, A Unified View of Simple EAs- A Common Framework, Population Size.	**CO3**
IV	**UNIT IV:** Hybrid Soft Computing Techniques: Introduction, Neuro Fuzzy Hybrid Systems, Comparison of Fuzzy Systems with Neural Networks, Characteristics of Neuro-Fuzzy Hybrids, Classification of Neuro Fuzzy Hybrid Systems, Genetic Neuro-Hybrid Systems: Properties of Genetic Neuro Hybrid Systems.	**CO4**
V	**UNIT V:** Neuro Fuzzy Modeling: ANFIS: Adaptive Neuro Fuzzy Inference Systems: Introduction, ANFIS Architecture, Hybrid Learning Algorithm, Learning Methods that cross fertilize ANFIS and RBFN, ANFIS as a Universal Approximator. Co-active Neuro Fuzzy Modeling: Towards Generalized ANFIS: Introduction, Framework, Neuron Functions for Adaptive networks, Neuro Fuzzy Spectrum.	**CO5**

Text Books:

1. Principles of soft computing S.N.Sivanandam, S.N.Deepa 3rd Wiley India 2019
2. Neuro-Fuzzy and Soft Computing J.S.R.Jang, C.T. Sun, E.Mizutani 1st Pearson Education 2004
3. Neural Networks, Fuzzy Logic and Genetic Algorithm: Synthesis and Applications S. Rajsekaran, G.A. Vijayalakshmi Pai 1st Prentice Hall of India 2003
4. Evolutionary Computation: A Unified Approach De Jong 1st MIT Press 2006

Reference Books:

1. Soft Computing & Intelligent Systems: Theory & Applications N. K. Sinha, M. M. Gupta 1st Academic Press 2009
2. Computational Intelligence: Concepts to Implementation R. Eberhart, Y. Shi 1st Morgan Kaufman 2007
3. Fuzzy Logic with engineering applications Timothy J. Ross 3rd Wiley India 2011
4. Computational Intelligence: An Introduction Andries P. Engelbrecht 2nd Wiley India 2009

E-Resources:

1. https://www.acad emia.edu/354761 56/Fundamentals _of_Soft_Computi ng
2. https://www.acad emia.edu/143572 4/Fuzzy_Logic_an d_Neural_Networ ks_by_Chennakes ava_R_Alavala
3. https://onlinecourses.nptel.ac.in/n
4. https://onlinecourses.nptel.ac.in/n oc22_cs54/preview

Course Code- Cloud Computing
(Professional Elective-II)

Lecture – Tutorial-Practical::	3-0-0	**Internal Marks:**	30
Credits:	3	**External Marks:**	70

Prerequisites: Students should have knowledge in

Computer Networks, **Operating Systems, Database Management Systems**, **Software Engineering**, **Programming Fundamentals**

Course Objectives:

The main objectives of the course are to

- **Explain and comprehend** fundamental cloud computing concepts including service models, deployment models, and enabling technologies.
- **Implement and deploy** cloud-based solutions using virtualization technologies, container platforms, and cloud service providers.
- **Examine and differentiate** between various cloud architectures by analyzing distributed computing patterns and virtualization techniques.
- **Assess and critique** cloud computing challenges including security frameworks, scalability requirements, and cost economics.
- **Design and develop** comprehensive cloud computing architectures integrating advanced concepts like serverless computing and IoT integration.

Course Outcomes:		**BTL LEVEL**
Upon successful completion of the course, the student will be able to:		
CO1	Compare and contrast different cloud service models (IaaS, PaaS, SaaS) and deployment models to select appropriate solutions.	**L4**
CO2	Apply parallel and distributed computing concepts including SOA, web services, and RPC to design scalable cloud applications.	**L3**
CO3	Implement virtualization technologies and container platforms (Docker, Kubernetes) to deploy and orchestrate cloud applications.	**L3**
CO4	Evaluate cloud computing challenges including security frameworks, economic models, and scalability issues to determine risk mitigation strategies.	**L5**
CO5	Create serverless computing architectures and IoT cloud integration solutions using FaaS platforms and edge computing principles.	**L6**

Contribution of Course Outcomes towards the achievement of Program Outcomes (1 – Low, 2-Medium, 3 – High)

CO	**PO1**	**PO 2**	**PO3**	**PO4**	**PO 5**	**PO6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PS O1**	**PS O2**	**PSO3**
CO1	3	3	2	2	-	-	-	-	-	2	-	-	3	2	2
CO2	3	3	3	3	2	-	-	-	2	-	-	2	2	-	-
CO3	3	-	-	3	-	2	-	2	2	2	2	-	2	3	3
CO4	-	3	3	-	2	2	2	-	-	-	2	3	3	2	2
CO5	3	-	3	3	3	2	2	2	2	2	2	2	2	-	3

Syllabus		
UnitNo	**Contents**	**MappedCO**
I	**UNIT -I**: Introduction to Cloud Computing Fundamentals Cloud computing at a glance, defining a cloud, cloud computing reference model, types of services (IaaS, PaaS, SaaS), cloud deployment models (public, private, hybrid), utility computing, cloud computing characteristics and benefits, cloud service providers (Amazon Web Services, Microsoft Azure, Google AppEngine).	**CO1**
II	**UNIT-II**: Cloud Enabling Technologies Ubiquitous Internet, parallel and distributed computing, elements of parallel computing, hardware architectures for parallel computing (SISD, SIMD, MISD, MIMD), elements of distributed computing, Inter-process communication, technologies for distributed computing, remote procedure calls (RPC), service-oriented architecture (SOA), Web services, virtualization.	**CO2**
III	**UNIT-III**: Virtualization and Containers Characteristics of virtualized environments, taxonomy of virtualization techniques, virtualization and cloud Computing, pros and cons of virtualization, technology examples (XEN, VMware), building blocks of containers, container platforms (LXC, Docker), container orchestration, Docker Swarm and Kubernetes, public cloud VM (e.g. Amazon EC2) and container (e.g. Amazon Elastic Container Service) offerings.	**CO3**
IV	**UNIT-IV**: Cloud computing challenges Economics of the cloud, cloud interoperability and standards, scalability and fault tolerance, energy efficiency in clouds, federated clouds, cloud computing security, fundamentals of computer security, cloud security architecture, cloud shared responsibility model, security in cloud deployment models.	**CO4**
V	**UNIT -V**: Advanced concepts in cloud computing Serverless computing, Function-as-a-Service, serverless computing architecture, public cloud (e.g. AWS Lambda) and open-source (e.g. OpenFaaS) serverless platforms, Internet of Things (IoT), applications, cloud-centric IoT and layers, edge and fog computing, DevOps, infrastructure-as-code, quantum cloud computing.	**CO5**

Text Books:

1. Sunit Belapure Nina Godbole "Cyber Security: Understanding Cyber Crimes, Computer Forensics and Legal Perspectives", WILEY, 2011.
2. Nelson Phillips and Enfinger Steuart, "Computer Forensics and Investigations", Cengage Learning, New Delhi, 2009.

Reference Books:

1. Michael T. Simpson, Kent Backman and James E. Corley, "Hands on Ethical Hacking and Network Defence", Cengage, 2019.
2. Computer Forensics, Computer Crime Investigation by John R. Vacca, Firewall Media, New Delhi.
3. Alfred Basta, Nadine Basta,Mary Brown and Ravinder Kumar "Cyber Security and Cyber Laws" , Cengage,2018.

E-Resources:

1. https://onlinecourses.swayam2.ac.in/nou19_cs08/preview
2. CERT-In Guidelines- http://www.cert-in.org.in/
3. https://www.coursera.org/learn/introduction-cybersecurity-cyber-attacks [Online Course]
4. https://computersecurity.stanford.edu/free-online-videos [Free Online Videos]
5. Nickolai Zeldovich. 6.858 Computer Systems Security. Fall 2014. Massachusetts Institute of Technology: MIT OpenCourseWare, https://ocw.mit.edu License: Creative Commons BY-NC-SA.

Course Code- Agentic Systems and Generative AI
(Professional Elective-II)

Lecture – Tutorial-Practical::	3-0-0	**Internal Marks:**	30
Credits:	3	**External Marks:**	70

Prerequisites: Students should have knowledge in

Machine Learning Fundamentals, Programming Skills, Mathematics and Statistics, Data Structures and Algorithms

Course Objectives:
The main objectives of the course are to

- Understand the theoretical foundations of intelligent agents and agentic architectures.
- Explore the principles and components of generative AI models such as GANs, VAEs, and diffusion models.
- Analyze the architecture and capabilities of Large Language Models (LLMs) and their agentic behaviors.
- Design and implement basic generative AI and agent-based systems for real-world tasks.
- Evaluate the ethical, societal, and security implications of deploying generative and agentic AI systems.

Course Outcomes:		**BTL LEVEL**
Upon successful completion of the course, the student will be able to:		
CO1	Analyze different types of intelligent agents and design appropriate agent architectures based on environmental characteristics and real-world application requirements.	**L4**
CO2	Apply generative AI concepts to distinguish between generative and discriminative models, understand latent space representations, and evaluate ethical implications including bias, misinformation, and plagiarism in AI-generated content.	**L3**
CO3	Implement and compare various generative models including autoencoders, VAEs, GANs and diffusion models.	**L3**

CO4	Create LLM-based agentic systems using transformer architectures with prompt engineering, fine-tuning techniques, and integration of tool use, memory, and planning through frameworks	**L6**
CO5	Evaluate multi-agent system coordination, apply game theory principles for agent interactions, implement agent-based simulations, and assess security, bias, and regulatory aspects of AI systems for responsible deployment.	**L5**

Contribution of Course Outcomes towards the achievement of Program Outcomes (1 – Low, 2-Medium, 3 – High)

CO	PO1	PO 2	PO3	PO4	PO 5	PO6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2	PSO3
CO1	2	3	-	2	2	2	-	2	-	2	-	3	3	2	2
CO2	2	3	3	-	2	-	-	-	-	2	-	2	-	3	3
CO3	3	3	-	-	3	-	-	-	-	2	-	2	3	3	-
CO4	2	3	3	2	3	2	-	2	-	3	2	2	3	3	2
CO5	3	3	3	3	3	2	-	2	2	3	3	2	3	-	3

Syllabus		
UnitNo	**Contents**	**MappedCO**
I	UNIT I: Introduction to AI and Agentic Systems, Types of agents: reflex, goal-based, utility-based, learning agents, Agent architectures: Reactive, Deliberative, Hybrid, Belief-Desire-Intention (BDI) model, Agent environments: deterministic, stochastic, fully and partially observable, Real-world applications of agents in robotics and software systems.	**CO1**
II	UNIT II: Introduction to Generative AI, Generative vs. discriminative models, Applications of generative AI: text, image, code, audio, Latent space representation and sampling, Deep learning foundations for generative tasks, Ethical concerns in generative AI: bias, misinformation, plagiarism.	**CO2**
III	UNIT III: Autoencoders and Variational Autoencoders (VAE), GANs: architecture and training dynamics, Types of GANs: DCGAN, CycleGAN, StyleGAN, Introduction to Diffusion Models, DeepFake generation and detection, Hands-on implementation of generative models.	**CO3**
IV	UNIT IV: Transformer architecture and self-attention mechanism, Large Language Models (LLMs): BERT, GPT, T5, Prompt engineering and fine-tuning techniques, LLMs as agents: AutoGPT, BabyAGI, LangChain, Integration of tool use, memory, and planning in agentic LLMs, Building intelligent assistants using LLM APIs.	**CO4**
V	UNIT V: UNIT V: Multi-agent systems: interaction, cooperation, coordination, Basics of game theory for intelligent agents, Agent-based simulation frameworks: NetLogo, Mesa, Security and bias in agentic and generative systems, AI alignment, safety, and	**CO5**

	control mechanisms, Societal implications and regulatory aspects of generative AI and autonomous agents.	

Text Books:

1. Denis Rothman, "Transformers for Natural Language Processing and Computer Vision", Third Edition , Packt Books, 2024

Reference Books:

1. Altaf Rehmani, "Generative AI for Everyone", BlueRose One, 2024.
2. Stuart Russell and Peter Norvig, *Artificial Intelligence: A Modern Approach*, Fourth Edition, Pearson, 2020.
   ISBN: 978-0134610993
3. Ian Goodfellow, Yoshua Bengio, and Aaron Courville, *Deep Learning*, MIT Press, 2016.
   ISBN: 978-0262035613
4. David Foster, *Generative Deep Learning: Teaching Machines to Paint, Write, Compose, and Play*, Second Edition, O'Reilly Media, 2023.
   ISBN: 978-1098134174
5. Francesco Ricci, Lior Rokach, and Bracha Shapira, *Recommender Systems Handbook*, Second Edition, Springer, 2015.
   ISBN: 978-1489976376
6. Thomas G. Dietterich, *Machine Learning*, Draft Edition, Available via online open educational resources.
7. Sebastian Raschka and Vahid Mirjalili, *Python Machine Learning: Machine Learning and Deep Learning with Python, scikit-learn, and TensorFlow 2*, Third Edition, Packt Publishing, 2019.
   ISBN: 978-1789955750
8. Denis Rothman, *Transformers for Natural Language Processing and Computer Vision*, Third Edition, Packt Publishing, 2024.
   ISBN: 978-1805124400

Online Resources:

https://www.analyticsvidhya.com/blog/2024/01/ultimate-list-of-generative-ai-resources/

Course Code- BLOCKCHAIN TECHNOLOGY

Lecture – Tutorial- Practical::	3-0-0				**Internal Marks:**	30
Credits:	3				**External Marks:**	70

Prerequisites:

1. Knowledge in security and applied cryptography.
2. Knowledge in distributed databases.

Course Objectives:

- To Introduce block chain technology and Crypto currency
- Discuss the fundamental ideas of Bit coin MOOCs.
- Identify the different components of the block chain process.
- Recognize the differences between block chain and bit coin and other crypto currencies.

Course Outcomes:

Upon successful completion of the course, the student will be able to:

CO1	Analyze blockchain fundamentals and compare different blockchain categories with their network architectures.	L4
CO2	Apply Bitcoin blockchain operations and implement various consensus mechanisms (PoW, PoS, BFT)	L3
CO3	Evaluate blockchain applications in digital identity verification and digital art environments.	L5
CO4	Analyze blockchain applications in scientific research and evaluate tokenization strategies for public adoption.	L4
CO5	Evaluate technical challenges, business models, and assess government regulations in blockchain implementation.	L5

Contribution of Course Outcomes towards achievement of Program Outcomes (1 – Low, 2-Medium, 3 – High)

	PO 1	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PSO1**	**PSO2**	**PSO 3**
CO1	3	3	2	-	-	-	-	-	-	-	-	3	2	3	-
CO2	3	3	3	2	2	2	-	-	2	2	-	3	2	3	-
CO3	**2**	**-**	**-**	**-**	**-**	**-**	**-**	**3**	**-**	**3**	**-**	**-**	**2**	**-**	**-**
CO4	3	2	3	2	-	-	-	-	-	-	-	3	3	2	3
CO5	3	2	3	2	-	-	-	-	-	-	-	3	3	2	3

UNIT I :

Introduction Introduction to Block chain – History, Definition, Distributed Ledger, Blockchain Categories – Public, Private, Consortium, Blockchain Network and Nodes, Peer-to-Peer Network, Mining Mechanism, Generic elements of Blockchain, Features of Blockchain, and Types of Blockchain.

UNIT II:

Operation of Bitcoin Blockchain, Blockchain Architecture – Block, Hash, Distributer P2P, Structure of Blockchain- Consensus mechanism: Proof of Work (PoW), Proof of Stake (PoS), Byzantine Fault Tolerance (BFT), Proof of Authority (PoA) and Proof of Elapsed Time (PoET)

UNIT III:

Extensibility of Blockchain concepts, Digital Identity verification, Block chain Neutrality, Digital art, Blockchain Environment
UNIT IV:
Blockchain Science: Gridcoin, Foldingcoin, Blockchain Genomics, Bitcoin MOOCs.Currency, Token, Tokenizing, Campuscoin, Coin drop as a strategy for Public adoption.Regulations
UNIT V:
Currency Multiplicity, Demurrage currency. Technical challenges, Business model challenges, Scandals and Public perception, Government Regulations
TEXT BOOKS:
1. Blockchain Blue print for Economy by Melanie Swan
REFERENCE BOOKS:
1. Blockchain Basics: A Non-Technical Introduction in 25 Steps 1st Edition, by Daniel Drescher
E-RESOURCES:
1. **Blockchain | Tools, Publications & Resources (ala.org)** 2. **Blockchain Learning Hub | UNICEF Office of Innovation**

Course Code- Natural Language Processing
(Professional Elective-III)

Lecture – Tutorial-Practical::	3-0-0	**Internal Marks:**	30
Credits:	3	**External Marks:**	70

Prerequisites: Students should have knowledge in

Programming Fundamentals, Mathematics and Statistics, Formal Language Theory, Data Structures and Algorithms, Computer Science Fundamentals

Course Objectives:
The main objectives of the course are to

- Master language modeling, regex, finite-state automata, morphological analysis, and spelling correction
- Implement N-gram models, smoothing techniques, POS tagging, and HMMs for lexical analysis
- Compare CFGs, dependency parsing, probabilistic parsing, and feature structures for grammatical analysis
- Apply logic representations, word sense disambiguation, thematic roles, and selectional restrictions for meaning extraction

- Build complete NLP applications incorporating discourse analysis, anaphora/coreference resolution, and lexical resources

Course Outcomes:		**BTL LEVEL**
Upon successful completion of the course, the student will be able to:		
CO1	**Apply** language modeling techniques including grammar-based and statistical approaches with finite-state automata for morphological analysis.	**L3**
CO2	**Implement** N-gram models with smoothing techniques and part-of-speech tagging systems using rule-based, stochastic, and HMM approaches.	**L3**
CO3	**Analyze** syntactic structures using context-free grammars, dependency parsing, and probabilistic parsing algorithms including CYK parser.	**L4**
CO4	**Evaluate** semantic representation systems using first-order logic and implement word sense disambiguation techniques with distributional similarity measures.	**L5**
CO5	**Design** discourse analysis systems implementing anaphora and coreference resolution algorithms with lexical resources integration.	**L6**

Contribution of Course Outcomes towards the achievement of Program Outcomes (1 – Low, 2-Medium, 3 – High)

CO	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**	**PSO1**	**PSO2**	**PSO3**
CO1	3	3	3	2	-	2	-	-	-	-	-	2	3	2	2
CO2	3	3	-	-	-	-	-	2	-	2	-	-	3	3	-
CO3	3	-	3	3	3	2	-	-	2	2	2	2	3	2	3
CO4	-	3	-	-	3	-	2	2	-	-	2	2	-	3	2
CO5	-	-	3	3	-	2	2	2	2	-	2	-	3	-	-

Syllabus		
UnitNo	**Contents**	**MappedCO**
I	**UNIT I:** **INTRODUCTION:** Origins and challenges of NLP – Language Modeling: Grammar-based LM, Statistical LM – Regular Expressions, Finite-State Automata – English Morphology, Transducers for lexicon and rules, Tokenization, Detecting and Correcting Spelling Errors, Minimum Edit Distance.	**CO1**
II	**UNIT II:** **WORD LEVEL ANALYSIS:** Unsmoothed N-grams, Evaluating N-grams, Smoothing, Interpolation and Backoff – Word Classes, Part- of-Speech Tagging, Rule-based, Stochastic and Transformation-based tagging, Issues in PoS tagging – Hidden Markov and Maximum Entropy models.	**CO2**

III	**UNIT III:** **SYNTACTIC ANALYSIS**: Context-Free Grammars, Grammar rules for English, Treebanks, Normal Forms for grammar – Dependency Grammar – Syntactic Parsing, Ambiguity, Dynamic Programming parsing – Shallow parsing Probabilistic CFG, Probabilistic CYK, Probabilistic Lexicalized CFGs – Feature structures, Unification of feature structures.	**CO3**
IV	**UNIT IV:** **SEMANTICS AND PRAGMATICS:** Requirements for representation, First-Order Logic, Description Logics – Syntax-Driven Semantic analysis, Semantic attachments – Word Senses, Relations between Senses, Thematic Roles, selectional restrictions – Word Sense Disambiguation, WSD using Supervised, Dictionary & Thesaurus, Bootstrapping methods – Word Similarity using Thesaurus and Distributional methods.	**CO4**
V	**UNIT V:** **DISCOURSE ANALYSIS AND LEXICAL RESOURCES:** Discourse segmentation, Coherence – Reference Phenomena, Anaphora Resolution using Hobbs and Centering Algorithm – Coreference Resolution – Resources: Porter Stemmer, Lemmatizer, Penn Treebank, Brill's Tagger, WordNet, PropBank, FrameNet, Brown Corpus, British National Corpus (BNC).	**CO5**

Text Books:

1. Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics and Speech, 2nd Edition, Daniel Jurafsky, James H. Martin -Pearson Publication,2014.
2. Natural Language Processing with Python, First Edition, Steven Bird, Ewan Klein and Edward Loper, OReilly Media,2009.

Reference Books:

1. Language Processing with Java and Ling Pipe Cookbook, 1st Edition, Breck Baldwin, Atlantic Publisher, 2015.
2. Natural Language Processing with Java, 2nd Edition, Richard M Reese, OReilly Media,2015.
3. Handbook of Natural Language Processing, Second, Nitin Indurkhya and Fred J. Damerau, Chapman and Hall/CRC Press, 2010.Edition
4. Natural Language Processing and Information Retrieval, 3rd Edition, Tanveer Siddiqui, U.S. Tiwary, Oxford University Press,2008.

e-Resource:

https://onlinecourses.nptel.ac.in/noc23_cs45/preview

Course Code-<u>Big Data Analytics</u>
(Professional Elective-III)

Lecture – Tutorial-Practical::	3-0-0	**Internal Marks:**	30
Credits:	3	**External Marks:**	70

Prerequisites: Students should have knowledge in

1. **Programming Fundamentals**

2. **Data Structures and Algorithms**
3. **Computer Networks**
4. Software Engineering
5. Mathematics and Statistics

Course Objectives:

The main objectives of the course is to

- **Understand and explain** the fundamentals of big data analytics including key trends, unstructured data processing, industry applications, and big data technologies to establish comprehensive knowledge of the big data ecosystem and its business applications.
- **Apply and implement** NoSQL database technologies including aggregate data models, key-value stores, document databases, graph databases, and distributed systems concepts like sharding and replication to manage large-scale data storage solutions.
- **Analyze and compare** Hadoop ecosystem components including HDFS architecture, MapReduce processing, Hive data warehousing, and data formats to determine optimal approaches for big data processing and analytics workflows.
- **Evaluate and assess** Apache Spark technologies including RDD operations, DataFrame transformations, Catalyst optimizer, streaming processing, and performance tuning strategies to determine efficiency and scalability of big data processing solutions.
- **Design and develop** comprehensive big data analytics systems integrating streaming processing, event-time analysis, structured streaming, and deployment strategies to create scalable real-time analytics applications for business intelligence and decision making.

Course Outcomes:		**BTL LEVEL**
Upon successful completion of the course, the student will be able to:		
CO1	Apply big data concepts and technologies to analyze industry use cases including web analytics, fraud detection, risk management, healthcare analytics, and algorithmic trading while implementing Hadoop and cloud-based big data solutions.	**L3**
CO2	Implement NoSQL database systems including aggregate data models, key-value stores, document databases, and graph databases while demonstrating distribution models, consistency mechanisms, and Cassandra database operations.	**L3**
CO3	Analyze and implement Hadoop ecosystem components including HDFS architecture, MapReduce processing, data integrity mechanisms, and Hive data warehousing with HiveQL operations, partitioning, and optimization techniques.	**L4**
CO4	Evaluate Apache Spark architecture and implement RDD operations, DataFrame transformations, Catalyst optimizer, deployment strategies, and performance tuning techniques for distributed big data processing applications.	**L5**
CO5	Design and develop streaming analytics systems using Spark streaming fundamentals, event-time processing, stateful processing, windowing operations, and structured streaming for real-time big data analytics applications.	**L6**

Contribution of Course Outcomes towards the achievement of Program Outcomes (1 – Low, 2-Medium, 3 – High)

CO	**PO1**	**PO 2**	**PO3**	**PO4**	**PO 5**	**PO6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PS O1**	**PS O2**	**PSO3**
CO1	3	-	-	-	-	-	-	-	2	-	2	2	3	-	-
CO2	3	3	3	2	3	2	-	-	-	2	-	-	2	3	3
CO3	-	2	-	3	3	-	-	-	2	-	2	-	-	-	-
CO4	2	3	3	-	-	2	-	2	-	-	2	2	3	2	3
CO5	3	3	3	3	-	2	2	2	-	2-	2	2	-	3	-

Syllabus		
UnitNo	**Contents**	**MappedCO**
I	**UNIT I:** **INTRODUCTION:** Origins and challenges of NLP – Language Modeling: Grammar-based LM, Statistical LM – Regular Expressions, Finite-State Automata – English Morphology, Transducers for lexicon and rules, Tokenization, Detecting and Correcting Spelling Errors, Minimum Edit Distance.	**CO1**
II	**UNIT II:** **WORD LEVEL ANALYSIS:** Unsmoothed N-grams, Evaluating N-grams, Smoothing, Interpolation and Backoff – Word Classes, Part- of-Speech Tagging, Rule-based, Stochastic and Transformation-based tagging, Issues in PoS tagging – Hidden Markov and Maximum Entropy models.	**CO2**
III	**UNIT III:** **SYNTACTIC ANALYSIS**: Context-Free Grammars, Grammar rules for English, Treebanks, Normal Forms for grammar – Dependency Grammar – Syntactic Parsing, Ambiguity, Dynamic Programming parsing – Shallow parsing Probabilistic CFG, Probabilistic CYK, Probabilistic Lexicalized CFGs – Feature structures, Unification of feature structures.	**CO3**
IV	**UNIT IV:** **SEMANTICS AND PRAGMATICS:** Requirements for representation, First-Order Logic, Description Logics – Syntax-Driven Semantic analysis, Semantic attachments – Word Senses, Relations between Senses, Thematic Roles, selectional restrictions – Word Sense Disambiguation, WSD using Supervised, Dictionary & Thesaurus, Bootstrapping methods – Word Similarity using Thesaurus and Distributional methods.	**CO4**
V	**UNIT V:** **DISCOURSE ANALYSIS AND LEXICAL RESOURCES:** Discourse segmentation, Coherence – Reference Phenomena, Anaphora Resolution using Hobbs and Centering Algorithm – Coreference Resolution – Resources: Porter Stemmer, Lemmatizer, Penn Treebank, Brill's Tagger, WordNet, PropBank, FrameNet, Brown Corpus, British National Corpus (BNC).	**CO5**

Text Books:

1. Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics and Speech, 2nd Edition, Daniel Jurafsky, James H. Martin -Pearson Publication,2014.
2. Natural Language Processing with Python, First Edition, Steven Bird, Ewan Klein and Edward Loper, OReilly Media,2009.

Reference Books:

1. Language Processing with Java and Ling Pipe Cookbook, 1st Edition, Breck Baldwin, Atlantic Publisher, 2015.
2. Natural Language Processing with Java, 2nd Edition, Richard M Reese, OReilly Media,2015.

3. Handbook of Natural Language Processing, Second, Nitin Indurkhya and Fred J. Damerau, Chapman and Hall/CRC Press, 2010.Edition
4. Natural Language Processing and Information Retrieval, 3rdEdition, Tanveer Siddiqui, U.S. Tiwary, Oxford University Press,2008.

e-Resource:
https://onlinecourses.nptel.ac.in/noc23_cs45/preview

Course Code-Distributed Operating Systems
(Professional Elective-III)

Lecture – Tutorial-Practical::	3-0-0	**Internal Marks:**	30
Credits:	3	**External Marks:**	70

Prerequisites: Students should have knowledge in

1. **Programming Fundamentals**
2. **Data Structures and Algorithms**
3. **Computer Networks**
4. System Programming
5. Operating Systems

Course Objectives:
The main objectives of the course are to

- **Understand and explain** Distributed computing fundamentals and system models
- **Apply and implement** Message passing and RPC implementation techniques
- **Analyze and compare** DSM systems and synchronization mechanisms
- **Evaluate and assess** Resource management and performance optimization
- **Design and develop** Complete distributed file systems with fault tolerance

Course Outcomes:		**BTL LEVEL**
Upon successful completion of the course, the student will be able to:		
CO1	Apply distributed computing system models and implement message passing systems with synchronization, buffering, process addressing, and failure handling mechanisms while demonstrating IPC mechanisms in distributed environments.	**L3**
CO2	Implement Remote Procedure Call (RPC) mechanisms including stub generation, marshaling techniques, server management, parameter-passing semantics, and security features while optimizing RPC performance in heterogeneous environments.	**L3**
CO3	Analyze distributed shared memory (DSM) architectures and implement synchronization algorithms including clock synchronization, event ordering, mutual exclusion, deadlock prevention, and election algorithms for distributed system coordination.	**L4**
CO4	Evaluate resource management strategies including global scheduling algorithms, task assignment approaches, load balancing techniques, and process migration mechanisms to optimize distributed system performance and resource utilization.	**L5**

CO5	Design distributed file systems with appropriate file models, caching schemes, replication strategies, fault tolerance mechanisms, and atomic transaction support to ensure data consistency and system reliability in distributed environments.	**L6**

Contribution of Course Outcomes towards the achievement of Program Outcomes (1 – Low, 2-Medium, 3 – High)

CO	PO1	PO 2	PO3	PO4	PO 5	PO6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2	PSO3
CO1	3	-	3	2	-	-	-	-	-	2	-	2	3	3	2
CO2	3	3	3	3	-	-	-	2	2	-	2	-	3	-	3
CO3	-	3	3	-	3	2	2	2	-	-	2	-	3	3	-
CO4	3	-	-	3	-	-	2	-	2	2	2	2	-	3	2
CO5	3	3	3	3	-	2	2	2	2	-	2	-	3	3	3

Syllabus		
UnitNo	**Contents**	**MappedCO**
I	**Unit I:** **Fundamentals:** What is Distributed Computing Systems? Evolution of Distributed Computing System; Distributed Computing System Models; What is Distributed Operating System? Issues in Designing a Distributed Operating System; Introduction to Distributed ComputingEnvironment(DCE). **Message Passing:** Introduction, Desirable features of a Good Message Passing System, Issues in PC by Message Passing, Synchronization, Buffering, Multi-datagram Messages, Encoding and Decoding of Message Data, Process Addressing, Failure Handling, Group Communication, Case Study: 4.3 BSD UNIX IPC Mechanism.	**CO1**
II	**Unit II:Remote Procedure Calls:** Introduction, The RPC Model, Transparency of RPC, Implementing RPC Mechanism, Stub Generation, RPC Messages, Marshaling Arguments and Results, Server Management, Parameter-Passing Semantics, Call Semantics, Communication Protocols for RPCs, Complicated RPCs, Client-Server Binding, Exception Handling, Security, Some Special Types of RPCs, RPC in Heterogeneous Environments, Lightweight RPC, Optimization for Better Performance, Case Studies: Sun RPC	**CO2**
III	**Unit III: Distributed Shared Memory:** Introduction, General Architecture of DSM systems, Design and Implementation Issues of DSM, Granularity, Structure of Shared Memory Space, Consistency Models, Replacement Strategy, Thrashing, Other approaches to DSM, Heterogeneous DSM, Advantages of DSM. Synchronization: Introduction, Clock Synchronization, Event Ordering, Mutual Exclusion, Dead Lock, Election Algorithms	**CO3**

IV	**Unit IV:Resource Management:** Introduction, Desirable Features of a Good Global Scheduling Algorithm, Task Assignment Approach, Load – Balancing Approach, Load – Sharing Approach Process Management: Introduction, Process Migration, Threads.	**CO4**
V	**Unit V: Distributed File Systems:** Introduction, Desirable Features of a Good Distributed File System, File models, File–Accessing Models, File – Sharing Semantics, File – Caching Schemes, File Replication, Fault Tolerance, Atomic Transactions and Design Principles.	**CO5**

Text books

1. Pradeep. K. Sinha: Distributed Operating Systems: Concepts and Design, PHI, 2007.

Reference Books:

1. Andrew S. Tanenbaum: Distributed Operating Systems, Pearson Education, 2013.
2. Ajay D. Kshemkalyani and MukeshSinghal, Distributed Computing: Principles, Algorithms and Systems, Cambridge University Press, 2008
3. Sunita Mahajan, Seema Shan, " Distributed Computing", Oxford University Press,2015

e-Resource:

https://onlinecourses.nptel.ac.in/noc21_cs87/preview

Course Code- Deep Learning Lab

Lecture – Tutorial-Practical::	0-0-3	**Internal Marks:**	30
Credits:	1.5	**External Marks:**	70

Prerequisites: Machine Learning, Python Programming, Mathematics (Linear Algebra, Calculus), Data Structures and Algorithms, Probability and Statistics

Course Objectives:

The objectives of this course are to

- **Understand** deep learning basics including neural networks, activation functions, and backpropagation algorithms.
- **Implement** different neural network types (MLPs, CNNs, RNNs) using TensorFlow, Keras, and PyTorch frameworks.
- **Compare** various deep learning models for classification, regression, computer vision, and text processing tasks.
- **Evaluate** pre-trained models and transfer learning techniques to select the best approach for specific problems.
- **Create** complete deep learning applications by combining preprocessing, model training, and evaluation techniques.

Course Outcomes:	**BTL LEVEL**

Upon successful completion of the course, the student will be able to:

CO1	Implement multi-layer perceptron algorithms and neural network architectures to solve fundamental classification and regression problems including handwritten digit recognition and text classification using appropriate datasets and evaluation metrics.	L3
CO2	Apply convolutional neural network architectures for image classification tasks including building CNN models from scratch and implementing various CNN designs for computer vision applications.	L3
CO3	Choose and evaluate appropriate pre-trained models including VGG16 to solve real-time computer vision problems while demonstrating transfer learning techniques and model adaptation strategies for practical applications.	L5
CO4	Interpret and analyze results of different deep learning models by implementing advanced text processing techniques including one-hot encoding, word embeddings, and recurrent neural networks for natural language processing applications.	L4

Contribution of Course Outcomes towards the achievement of Program Outcomes (1 – Low, 2-Medium, 3 – High)

	PO1	**PO 2**	**PO3**	**PO4**	**PO 5**	**PO6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PS O1**	**PS O2**	**PSO3**
CO1	3	2	3	2	-	2	-	2	-	-	2	3	2	3	2
CO2	3	3	3	-	3	2	-	-	2	2	2	-	3	2	3
CO3	2	2	2	3	3	-	2	-	-	2	-	3	2	3	2
CO4	3	3	3	3	3	2	2	2	2	3	2	3	3	3	3

List of Experiments:

1. Implement multi-layer perceptron algorithm for MNIST Handwritten Digit Classification.
2. Design a neural network for classifying movie reviews (Binary Classification) using IMDB dataset.
3. Design a neural Network for classifying news wires (Multi class classification) using Reuters dataset.
4. Design a neural network for predicting house prices using Boston Housing Price dataset.
5. Build a Convolution Neural Network for MNIST Handwritten Digit Classification.
6. Build a Convolution Neural Network for simple image(dogs and Cats) Classification
7. Use a pre-trained convolution neural network (VGG16) for image classification.
8. Implement one hoten coding of words or characters.
9. Implement word embeddings for IMDB dataset.
10. Implement a Recurrent Neural Network for IMDB movie review classification problem.

Text Books:

1. Reza Zadehand Bharath Ram sundar, "Tensorflow for Deep Learning", O'Reilly publishers, 2018

References:

1. https://github.com/fchollet/deep-learning-with-python-notebooks

Course Code- Quantum Computing Lab

Lecture – Tutorial-Practical::	0-0-3	**Internal Marks:**	30
Credits:	1.5	**External Marks:**	70

Prerequisites: Linear Algebra and Matrix Theory, Probability and Statistics, Programming Fundamentals (Python), Data Structures and Algorithms, Computer Organization

Course Objectives:

The objectives of this course are to

- **Understand** quantum computing fundamentals including quantum gates, qubits, superposition, entanglement, and quantum circuit design principles.
- **Implement** fundamental quantum algorithms including Deutsch's algorithm, Deutsch-Jozsa algorithm, Shor's factorization, and Grover's search using quantum programming frameworks.
- **Analyze** quantum algorithm complexity and performance compared to classical algorithms for specific computational problems.
- **Evaluate** practical applications of quantum computing in cybersecurity, finance, optimization, and scientific computing domains.
- **Design** comprehensive case studies demonstrating quantum computing solutions for real-world problems in various application domains.

Course Outcomes:		**BTL LEVEL**
Upon successful completion of the course, the student will be able to:		
CO1	Implement fundamental quantum algorithms including Deutsch's algorithm and Deutsch-Jozsa algorithm to demonstrate quantum parallelism and oracle-based computation principles using quantum programming frameworks.	L3
CO2	Apply advanced quantum algorithms including Shor's factorization technique and Grover's search algorithm to solve complex computational problems demonstrating quantum speedup over classical approaches.	L3
CO3	Analyze and evaluate quantum encryption methods and cybersecurity applications through comprehensive case study development demonstrating practical quantum cryptography implementations.	L5
CO4	Design and develop case studies for quantum computing applications in financial modeling, traffic optimization, weather forecasting, and artificial intelligence domains to demonstrate practical quantum solutions for real-world problems.	L4

Contribution of Course Outcomes towards the achievement of Program Outcomes (1 – Low, 2-Medium, 3 – High)

CO	PO1	PO 2	PO3	PO4	PO 5	PO6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2	PSO3
CO1	3	3	3	3	3	-	-	-	-	2	2	3	3	-	3
CO2	3	-	3	3	3	2	2	2	2	-	2	-	3	3	3
CO3	-	3	2	-	2	3	-	-	-	3	-	3	3	2	-
CO4	3	3	-	3	3	3	3	2	2	3	2	3	-	3	3

List of Experiments:

1. Write a program to implement Deutsch's algorithm
2. Write a program to implement Deutsch's-Jozsa algorithm
3. Write a program to implement Shor factorization technique
4. Write a program to implement Grover search technique
5. Prepare case study for any suitable application on quantum encryption methods for Cyber
6. Prepare case study for any suitable application for Financial Modeling
7. Prepare case study for any suitable application for Traffic Optimization
8. Prepare case study for any suitable application for Weather Forecasting and Climate Change
9. Prepare case study for any suitable application for Artificial Intelligence

Text Books:

- Quantum Computation and Quantum Information, M A Nielsen and I L Chuang, Cambridge University Press.
- Principles of Quantum Computation and Information, Vol. I: Basic Concepts, Vol II: Basic Tools and Special Topics, World Scientific.
- An Introduction to Quantum Computing, Pittenger,
- An Introduction to Quantum Computing, P Kaye, R Laflamme and M Mosca.
- Linear Algebra and its Applications, G. Strang.
- Matrix Analysis, Bhatia.
- Quantum Computing without Magic by Zdzis_law Meglicki
- Quantum Computing Explained By DAVID McMAHON
- Quantum Computer Science By Marco Lanzagorta, Jeffrey Uhlmann

e-References:

Courses by: Andrew Childs, Ronald de Wolf, and John Watrous on quantum theory and computing

Course Code- Soft skills
(Skill Enhancement Course)

Lecture – Tutorial-Practical::	3-0-0	**Internal Marks:**	30
Credits:	3	**External Marks:**	70

Prerequisites: Students should have knowledge in

Basic English Language Skills, Communication Fundamentals, Computer Literacy, General Knowledge, Social Interaction Skills

Course Objectives:

The main objectives of the course is to

- To equip the students with the skills to effectively communicate in English
- To train the students in interview skills, group discussions and presentation skills
- To motivate the students to develop confidence
- To enhance the students' interpersonal skills
  To improve the students' writing skills

Course Outcomes:		**BTL LEVEL**
Upon successful completion of the course, the student will be able to:		
CO1	Develop analytical thinking, listening skills, and effective verbal/non-verbal communication including body language interpretation	**L3**
CO2	Implement anger, stress, and time management techniques while mastering social, business, telephone, and dining etiquette	**L3**
CO3	Demonstrate proficiency in grammar, letter writing, note-taking, minutes preparation, and email communication	**L3**
CO4	Execute job-oriented skills including group discussions, resume preparation, and interview techniques through mock sessions	**L3**
CO5	Analyze relationship types, factors, and accommodation strategies to build effective professional and personal networks	**L4**

Contribution of Course Outcomes towards the achievement of Program Outcomes (1 – Low, 2-Medium, 3 – High)

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	-	2	-	-	-	2	2	2	-	3	3	3	-	-	2
CO2	-	2	-	-	-	3	-	3	-	3	3	3	-	-	-
CO3	2	-	-	-	-	2	2	2	2	3	3	3	-	2	2
CO4	2	2	2	-	-	2	2	-	3	3	3	3	-	-	-
CO5	2	2	2	-	-	3	2	3	3	3	3	3	-	-	2

Syllabus		
UnitNo	**Contents**	**MappedCO**

I	**UNIT – I** **Analytical Thinking & Listening Skills:** Self-Introduction, Shaping Young Minds - A Talk by Azim Premji (Listening Activity), Self – Analysis, Developing Positive Attitude, Perception. **Communication Skills:** Verbal Communication; Non Verbal Communication (Body Language)	**CO1**
II	**UNIT – II** **Self-Management Skills:** Anger Management, Stress Management, Time Management, Six Thinking Hats, Team Building, Leadership Qualities **Etiquette:** Social Etiquette, Business Etiquette, Telephone Etiquette, Dining Etiquette	**CO2**
III	**UNIT – III** **Standard Operation Methods :** Basic Grammars, Tenses, Prepositions, Pronunciation, Letter Writing; Note Making, Note Taking, Minutes Preparation, Email & Letter Writing	**CO3**
IV	**UNIT-IV** **Job-Oriented Skills:** Group Discussion, Mock Group Discussions, Resume Preparation, Interview Skills, Mock Interviews	**CO4**
V	**UNIT-V** **Interpersonal relationships**: Introduction, Importance, Types, Uses, Factors affecting interpersonal relationships, Accommodating different styles, Consequences of interpersonal relationships	**CO5**

Text Books

S.No	Book Details
1	Barun K. Mitra, Personality Development and Soft Skills, Oxford University Press, 2011.
2	S.P. Dhanavel, English and Soft Skills, Orient Blackswan, 2010.

Reference Books

S.No	Book Details
1	R.S. Aggarwal, A Modern Approach to Verbal & Non-Verbal Reasoning, S. Chand & Company Ltd., 2018.
2	Raman, Meenakshi & Sharma, Sangeeta, Technical Communication Principles and Practice, Oxford University Press, 2011.
3	Dale Carnegie, How to Win Friends and Influence People, Simon & Schuster, 2009.
4	Stephen R. Covey, The 7 Habits of Highly Effective People, Free Press, 2004.

E-Resources

S.No	Resource
1	https://swayam-plus.swayam2.ac.in/courses/course-details?id=P_CAMBR_01
2	Coursera Soft Skills Courses - https://www.coursera.org
3	LinkedIn Learning Professional Development - https://www.linkedin.com/learning
4	TED Talks on Communication and Leadership - https://www.ted.com

III Year II Semester	Technical Paper Writing & IPR	L	T	P	C
		2	0	0	-

Course Objective : The course will explain the basic related to writing the technical reports and understanding the concepts related to formatting and structuring the report. This will help students to comprehend the concept of proofreading, proposals and practice

Unit I:

Introduction: An introduction to writing technical reports, technical sentences formation, using transitions to join sentences, Using tenses for technical writing.
Planning and Structuring: Planning the report, identifying reader(s), Voice, Formatting and structuring the report, Sections of a technical report, Minutes of meeting writing.

Unit II:
Drafting report and design issues: The use of drafts, Illustrations and graphics.
Final edits: Grammar, spelling, readability and writing in plain English: Writing in plain English, Jargon and final layout issues, Spelling, punctuation and Grammar, Padding, Paragraphs, Ambiguity.

Unit III:
Proofreading and summaries: Proofreading, summaries, Activities on summaries.
Presenting final reports: Printed presentation, Verbal presentation skills, Introduction to proposals and practice.

Unit IV: Using word processor:
Adding a Table of Contents, Updating the Table of Contents, Deleting the Table of Contents, Adding an Index, Creating an Outline, Adding Comments, Tracking Changes, Viewing Changes, Additions, and Comments, Accepting and Rejecting Changes , Working with Footnotes and Endnotes, Inserting citations and Bibliography, Comparing Documents, Combining Documents, Mark documents final and make them read only., Password protect Microsoft Word documents., Using Macros,

Unit V:
Nature of Intellectual Property: Patents, Designs, Trade and Copyright. Process of **Patenting and Development:** technological research, innovation, patenting, development. International Scenario: International cooperation on Intellectual Property

Text Books:

1. Kompal Bansal & Parshit Bansal, "Fundamentals of IPR for Beginner's", 1st Ed., BS Publications, 2016.
2. William S. Pfeiffer and Kaye A. Adkins, "Technical Communication: A Practical Approach", Pearson.
3. Ramappa,T., "Intellectual Property Rights Under WTO", 2nd Ed., S Chand, 2015.

Reference Books:

1. Adrian Wallwork , English for Writing Research Papers, Springer New York Dordrecht Heidelberg London, 2011.
2. Day R, How to Write and Publish a Scientific Paper, Cambridge University Press(2006)

E-resources:
1. https://www.udemy.com/course/reportwriting/
2. https://www.udemy.com/course/professional-business-english-and-technical-report-writing/
3. https://www.udemy.com/course/betterbusinesswriting/